

# Sphere Rigidity

[Earlier public Project view and PDF](#)

MATHLIBANNEX / PROGRESSIVE RESEARCH COMPANION

## The mathematical goal

If at least one of two real normed spaces is finite-dimensional, their unit spheres are isometric in the chord metric exactly when the ambient spaces are linearly isometric. Equivalently, a surjective unit-sphere isometry yields a linear isometry equivalence of the ambient spaces under that finite-dimensional hypothesis.

Exact source: [MathlibAnnex v0.4.0 Project entry](#) · [Source manifest](#)

75 approved Declaration Cards are selected in this private candidate. The existing 467-declaration Project graph and levels are preserved; route presets remain inactive pending exact terminal metadata.

## Scope

Finite-dimensional real normed spaces and surjective isometries between their unit spheres, with the theorem stated when at least one ambient space is finite-dimensional. The formalization includes radial extension, dimension transfer, determinant and Pluecker methods, analytic orientation tools, finite recovery, and ball-volume rigidity used in the proof architecture.

## Scope limits

This Research Companion describes exact formal source and dependency structure. It does not assert completed Declaration Cards, document correspondence, or public LFH review.

## Proof architecture

### R01 · Radial Extensions and Dimension

Radial extension of sphere isometries and transfer of finite-dimensional information. The route does not claim that the radial extension is linear or globally isometric; the recorded 3-Lipschitz bound is the source constant, not an optimality claim.

### R02 · Compact Convex Hulls and Support Faces

Compact hulls, support faces, and finite lexicographic selection. The raw sets need not themselves be convex, nonemptiness is explicit, and

16 selected declarations

finite-dimensional hypotheses appearing in the exact source are preserved.

27 selected declarations

### R03 · Determinant Methods

Maximal-minor and determinant estimates with fixed row-order and sign conventions. Ordered row embeddings are not conflated with unordered row sets, and displayed bounds are not silently reinterpreted as Euclidean operator-norm statements.

24 selected declarations

### R04 · Piola Identity and Null Lagrangians

Piola-type identities, null Lagrangians, determinant continuity, and boundary traces. The whole-space perturbation identity concerns the integral of the determinant difference; Strong  $L^n$  includes eventual integrability, and the trace used here is pointwise on the sphere.

33 selected declarations

### R05 · Weak Gradient Zero and A.E. Constancy

Weak gradient zero implies local and global almost-everywhere constancy, not pointwise constancy. Compact  $C^1$  test fields and divergence form the shared analytic base used by the orientation route.

14 selected declarations

### R06 · Orientation and Signed Jacobians

Orientation and signed Jacobians on connected domains. No single global sign is asserted on a disconnected domain, and the null-image statement retained here is the same-dimension version present in the exact source.

0 selected declarations

### R07 · Determinant-Maximizing Dual Frames

Determinant-maximizing dual frames and inverse bounds, including the zero-dimensional case. Mathlib providers are preferred for norming functionals; application-specific satellite-polynomial material remains provenance rather than a new public API claim.

61 selected declarations

### R08 · Sphere-Volume Rigidity

Sphere-volume and ball-image rigidity. Compactness and inclusion in the relevant ball are explicit; equality of volume alone is not used to infer equality of sets without the separate inclusion hypothesis.

7 selected declarations

### R09 · Determinant Factorization

A factorization route from proportional maximal minors, retained as a heavy-refactor proof seed rather than an independently completed tool. Scale, sign, row order, and nondegeneracy conditions remain explicit to prevent degenerate overgeneralization.

20 selected declarations

# Boundary Inputs

3439 exact Boundary Inputs: EXTERNAL\_COMPILED\_DECLARATION 2632, OMITTED\_INTERNAL\_NATIVE\_DECLARATION 807. Selected reachability is retained through every omitted internal declaration.

## ▼ Inspect Boundary Inputs

- `Real` — Compiled external provider used by selected Project declarations.
- `Nat` — Compiled external provider used by selected Project declarations.
- `Eq` — Compiled external provider used by selected Project declarations.
- `OfNat.ofNat` — Compiled external provider used by selected Project declarations.
- `Real.normedField` — Compiled external provider used by selected Project declarations.
- `NormedAddCommGroup.toSeminormedAddCommGroup` — Compiled external provider used by selected Project declarations.
- `Real.semiring` — Compiled external provider used by selected Project declarations.
- `Semiring.toNonAssocSemiring` — Compiled external provider used by selected Project declarations.
- `DFunLike.coe` — Compiled external provider used by selected Project declarations.
- `UniformSpace.toTopologicalSpace` — Compiled external provider used by selected Project declarations.
- `PseudoMetricSpace.toUniformSpace` — Compiled external provider used by selected Project declarations.
- `NormedAddCommGroup.toAddCommGroup` — Compiled external provider used by selected Project declarations.
- `id` — Compiled external provider used by selected Project declarations.
- `NormedSpace.toModule` — Compiled external provider used by selected Project declarations.
- `NormedAddCommGroup` — Compiled external provider used by selected Project declarations.

# Dependency-first reading route

Levels belong to this Project. Select a level or follow a relation to another declaration tile.

## Level 0 46 declarations

LEVEL 0 R05

### AEConstantOn

MathlibAnnex.AEConstantOn

Mathematical construction

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

#### USED BY IN THIS PROJECT

[targetJacobianSign\\_const\\_is\\_pm\\_one](#),  
[compactLocalization\\_aeConstant\\_inner](#),  
[aeConstantOn\\_of\\_countableCover](#)

#### ▼ Show 2 more

[aeConstants\\_eq\\_of\\_open\\_overlap](#), [ConstantRegion](#)

[Read exact source](#)

LEVEL 0 Project-wide support

### BiLipschitzOpenData

MathlibAnnex.BilipschitzOrientation.BiLipschitzOpenData

Mathematical structure

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

#### USED BY IN THIS PROJECT

[f\\_injective](#), [lipschitz\\_f](#), [lipschitz\\_g](#)

#### ▼ Show 3 more

[domainJacobianSign](#), [fderiv\\_lower\\_bound](#),  
[target\\_exceptional\\_null](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

Project-wide support

## volume\_image\_eq\_zero\_of\_lipschitzWith

MathlibAnnex.BilipschitzOrientation.volume\_image\_eq\_zero\_of\_lipschitzWith

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[target\\_exceptional\\_null](#), [radialExtension\\_integral\\_abs\\_det](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## CompactC1VectorField

MathlibAnnex.CompactC1VectorField

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[carrier](#), [contDiff](#), [hasCompactSupport](#)

▼ Show 1 more

[ofSupport](#)

[Read exact source](#)

LEVEL 0

R07

## Frame

MathlibAnnex.DeterminantFrame.Frame

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[coordinateFrame](#), [frameCoordinates](#), [frameMap](#)

▼ Show 4 more

[frameMatrix](#), [replaceRow](#), [unitFrameSet](#),  
[SatelliteConfiguration](#)

[Read exact source](#)

LEVEL 0

R07

## NearMaxInverseBound

MathlibAnnex.DeterminantFrame.NearMaxInverseBound

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[K\\_nonneg](#), [bound\\_sub](#), [exists\\_nearMaxInverseBound](#)

▼ Show 3 more

[CoefficientDetectsUnit](#),  
[frameCoordinates\\_mem\\_coefficientAnnulus](#),  
[modelDist\\_framePreimage\\_le](#)

[Read exact source](#)

LEVEL 0

R07

## coordinateEquiv

MathlibAnnex.DeterminantFrame.coordinateEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[bound\\_sub](#), [frameCoordinates\\_eq\\_mulVec](#),  
[functional\\_apply\\_eq\\_sum](#)

▼ Show 2 more

[inverseCoordinateMap](#), [rawCoordinateRow](#)

[Read exact source](#)

LEVEL 0

R07

## functionalCoordinates

MathlibAnnex.DeterminantFrame.functionalCoordinates

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[frameMatrix\\_replaceRow](#), [functional\\_apply\\_eq\\_sum](#)

[Read exact source](#)

LEVEL 0

R07

## unitRowSet

MathlibAnnex.DeterminantFrame.unitRowSet

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[unitRowSet\\_isCompact](#), [mem\\_unitRowSet](#), [unitFrameSet](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## EquivalentSeminorm

MathlibAnnex.EquivalentSeminorm

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[IsContraction](#), [LinearIsometryEquiv](#), [Space](#)

▼ Show 12 more

[closedUnitBall](#), [eq\\_zero\\_of\\_apply\\_eq\\_zero](#),  
[lipschitzWith\\_upper\\_of\\_model\\_bound](#),  
[ofContinuousLinearEquiv](#), [unitSphere](#),  
[AlmostIsometryMatch](#), [FinMapAlmostIsometric](#),  
[LimitCertificate](#), [LinearCertificate](#),  
[abs\\_eval\\_lower\\_of\\_norms\\_nearby](#),  
[frameCoordinates\\_norm\\_le\\_model](#), [satelliteRowsSet](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

Project-wide support

## norm\_apply\_le\_seminorm\_of\_lipschitz

MathlibAnnex.FDeriv.norm\_apply\_le\_seminorm\_of\_lipschitz

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[ae\\_norm\\_apply\\_le\\_seminorm\\_of\\_lipschitz](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

Project-wide support

## satelliteAmbientDim

MathlibAnnex.FiniteSup.Bridge.satelliteAmbientDim

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[configurationFinMap](#), [finMapBaseRow](#), [finMapSatelliteRow](#)

[Read exact source](#)

LEVEL 0

R05

## LocalAECConstantAt

MathlibAnnex.LocalAECConstantAt

Mathematical structure

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[exists\\_localAECConstantAt](#), [constantRegion\\_compl\\_isOpen](#)

[Read exact source](#)

LEVEL 0

R03

## MaximalMinorIndex

MathlibAnnex.Matrix.MaximalMinorIndex

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[ofOrderEmbedding](#), [orderedRows](#)

[Read exact source](#)

LEVEL 0

R09

## chartFactor

MathlibAnnex.Matrix.chartFactor

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[chartFactor\\_unique](#),  
[mul\\_chartFactor\\_eq\\_of\\_orientedMaximalMinorsProportional](#),  
[selectedSubmatrix\\_mul\\_chartFactor](#)

[Read exact source](#)

LEVEL 0

R09

## det\_smul\_vecMul\_nonsingInv\_eq\_updateRowDet

MathlibAnnex.Matrix.det\_smul\_vecMul\_nonsingInv\_eq\_updateRowDet

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[rowCoordinates\\_eq\\_of\\_orientedMaximalMinorsProportional](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

R09

## exists\_orderEmbedding\_perm\_of\_injective

MathlibAnnex.Matrix.exists\_orderEmbedding\_perm\_of\_injective

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#)

[Read exact source](#)

LEVEL 0

R09

## orientedMaximalMinor

MathlibAnnex.Matrix.orientedMaximalMinor

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[OrientedMaximalMinorsProportional](#), [chartFactor\\_unique](#), [orientedMaximalMinor\\_eq\\_of\\_orderEmbedding](#)

### ▼ Show 3 more

[orientedMaximalMinor\\_eq\\_zero\\_of\\_not\\_injective](#), [orientedMaximalMinor\\_permute](#), [selectedSubmatrix\\_mul\\_chartFactor](#)

[Read exact source](#)

LEVEL 0

R03

## rowL1Norm

MathlibAnnex.Matrix.rowL1Norm

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[abs\\_det\\_le\\_prod\\_rowL1Norm](#)

[Read exact source](#)

LEVEL 0

R09

## submatrix\_update\_rowTuple

MathlibAnnex.Matrix.submatrix\_update\_rowTuple

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[rowCoordinates\\_eq\\_of\\_orientedMaximalMinorsProportional](#)

[Read exact source](#)

LEVEL 0

R04

## StrongLnOperatorField

MathlibAnnex.MeasureTheory.StrongLnOperatorField

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[tendsto\\_integral\\_det\\_of\\_strongLn](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## lipschitz\_fderiv\_memLpOn\_compact

MathlibAnnex.Mollification.lipschitz\_fderiv\_memLpOn\_compact

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#), [tendsto\\_integral\\_maximalMinor\\_mollify](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## standardMollifier

MathlibAnnex.Mollification.standardMollifier

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[standardMollifier\\_continuous](#),  
[standardMollifier\\_hasCompactSupport](#), [mollify](#)

[Read exact source](#)

LEVEL 0

R02

## maximizer

MathlibAnnex.NonemptyCompacts.maximizer

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[le\\_maximizer](#), [maxValue](#), [maximizer\\_mem](#)

[Read exact source](#)

LEVEL 0

R04

## basisRow

MathlibAnnex.Piola.basisRow

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[basisRow\\_apply](#), [cofactorRow](#), [hessianCoordinate](#)

#### ▼ Show 3 more

[singleOutputPerturb](#), [sum\\_smul\\_basisRow](#),  
[twoRowReplacement](#)

[Read exact source](#)

LEVEL 0

R04

## coordinateDivergence

MathlibAnnex.Piola.coordinateDivergence

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[coordinateDivergence\\_eq\\_zero\\_of\\_not\\_mem\\_tsupport](#)

[Read exact source](#)

LEVEL 0

R04

## det\_updateRow\_finset\_sum

MathlibAnnex.Piola.det\_updateRow\_finset\_sum

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[det\\_updateRow\\_eq\\_sum\\_mul\\_cofactorRow](#)

[Read exact source](#)

LEVEL 0

R04

## exists\_open\_box\_containing\_compact

MathlibAnnex.Piola.exists\_open\_box\_containing\_compact

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[integral\\_coordinateDivergence\\_eq\\_zero\\_of\\_contDiff\\_hasCompactSupport](#)

[Read exact source](#)

LEVEL 0

R04

## outputHybrid

MathlibAnnex.Piola.outputHybrid

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[outputHybrid\\_empty](#),  
[outputHybrid\\_insert\\_eq\\_singleOutputPerturb](#),  
[outputHybrid\\_univ](#)

[Read exact source](#)

LEVEL 0

R04

## sum\_symmetric\_mul\_antisymmetric\_eq\_zero

MathlibAnnex.Piola.sum\_symmetric\_mul\_antisymmetric\_eq\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[sum\\_hessian\\_twoRowReplacement\\_eq\\_zero](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## FiniteCoefficientNet

MathlibAnnex.Satellite.FiniteCoefficientNet

Mathematical structure

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[exists\\_net\\_preimage\\_close](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## SatelliteRows

MathlibAnnex.Satellite.SatelliteRows

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[SatelliteConfiguration](#), [satelliteRowsSet](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## abs\_affine\_endpoint\_rigidity

MathlibAnnex.Satellite.abs\_affine\_endpoint\_rigidity

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[absoluteMaximizer\\_satellite\\_norms\\_preimage](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## coefficientAnnulus

MathlibAnnex.Satellite.coefficientAnnulus

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[frameCoordinates\\_mem\\_coefficientAnnulus](#)

[Read exact source](#)

LEVEL 0

R08

## interior\_sdiff\_nonempty

MathlibAnnex.SeminormBall.interior\_sdiff\_nonempty

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[measure\\_lt](#)

[Read exact source](#)

LEVEL 0

R08

## map\_le

MathlibAnnex.SeminormBall.map\_le

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[map\\_le](#), [map\\_eq](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## isometryEquiv\_of\_linearIsometryEquiv

MathlibAnnex.Sphere.isometryEquivOfLinearIsometryEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[nonempty\\_isometryEquiv\\_iff\\_nonempty\\_linearIsometryEquiv\\_finiteDimensional](#),  
[nonempty\\_isometryEquiv\\_iff\\_nonempty\\_linearIsometryEquiv\\_of\\_finiteDimensional](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## normalizeToSphere

MathlibAnnex.Sphere.normalizeToSphere

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[coe\\_normalizeToSphere](#), [normalizeToSphere\\_pos\\_smul](#),  
[normalizeToSphere\\_unit](#)

▼ Show 1 more

[radialExtension](#)

[Read exact source](#)

LEVEL 0

R01

## scaled\_normalize\_dist\_le\_two

MathlibAnnex.Sphere.scaled\_normalize\_dist\_le\_two

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[radialExtension\\_lipschitz](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## LocalBallData

MathlibAnnex.WeakGradient.LocalBallData

Mathematical structure

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[carrier](#), [inner](#), [middle](#)

▼ Show 1 more

[exists\\_localBallData](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## compactLocalization

MathlibAnnex.WeakGradient.compactLocalization

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[compactLocalization\\_of\\_mem](#),  
[compactLocalization\\_of\\_not\\_mem](#),  
[compactLocalization\\_integrable](#)

#### ▼ Show 1 more

[weakIntegral\\_eq\\_localized\\_bumpIntegral](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## shrinkingBump

MathlibAnnex.WeakGradient.shrinkingBump

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[localMollification](#), [normalizedShrinkingBump](#),  
[shrinkingBump\\_rln](#)

#### ▼ Show 3 more

[shrinkingBump\\_rOut](#), [shrinkingBump\\_rOut\\_le\\_half](#),  
[shrinkingBump\\_rOut\\_tendsto\\_zero](#)

[Read exact source](#)

LEVEL 0

R02

## coordinate

MathlibAnnex.coordinate

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[allCoordinates](#)

[Read exact source](#)

LEVEL 0

Project-wide support

## divergence

MathlibAnnex.divergence

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[divergence\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#),  
[WeakDivergenceZero](#), [divergence\\_pi](#)

#### ▼ Show 1 more

[divergence\\_sub\\_smul](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

R02

## isCompact\_convexHull\_pi

MathlibAnnex.isCompact\_convexHull\_pi

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[derivativeAverage\\_mem\\_body](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

Project-wide support

## trace\_smulRight

MathlibAnnex.trace\_smulRight

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

### USED BY IN THIS PROJECT

[divergence\\_sub\\_smul](#)

[Read exact source](#)

**LEVEL 1**

Project-wide support

**f\_injective**

MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData.f\_injective

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#)

**USED BY IN THIS PROJECT**

[weak\\_piola](#)

[Read exact source](#)

**LEVEL 1**

Project-wide support

**lipschitz\_f**

MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData.lipschitz\_f

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#)

**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

**LEVEL 1**

Project-wide support

**lipschitz\_g**

MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData.lipschitz\_g

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#)

**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

**LEVEL 1**

Project-wide support

**domainJacobianSign**

MathlibAnnex.BilipschitzOrientation.domainJacobianSign

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#)

**USED BY IN THIS PROJECT**

[abs\\_domainJacobianSign](#), [det\\_eq\\_domainSign\\_mul\\_abs](#), [targetJacobianSign](#)

[Read exact source](#)

**LEVEL 1**

Project-wide support

**fderiv\_lower\_bound**

MathlibAnnex.BilipschitzOrientation.fderiv\_lower\_bound

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#)

**USED BY IN THIS PROJECT**

[fderiv\\_injective](#)

[Open Declaration Card](#) [Read exact source](#)

**LEVEL 1**

Project-wide support

**target\_exceptional\_null**

MathlibAnnex.BilipschitzOrientation.target\_exceptional\_null

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[BiLipschitzOpenData](#), [volume\\_image\\_eq\\_zero\\_of\\_lipschitzWith](#)

**USED BY IN THIS PROJECT**

[signed\\_area\\_transfer](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## carrier

MathlibAnnex.CompactC1VectorField.carrier

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CompactC1VectorField](#)

### USED BY IN THIS PROJECT

[carrier\\_ofSupport](#), [carrier\\_compact](#), [support\\_subset](#)

#### ▼ Show 3 more

[tsupport\\_subset](#), [WeakDivergenceZero](#),  
[translatedBumpField\\_carrier\\_subset](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## contDiff

MathlibAnnex.CompactC1VectorField.contDiff

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CompactC1VectorField](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#), [continuous](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## hasCompactSupport

MathlibAnnex.CompactC1VectorField.hasCompactSupport

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CompactC1VectorField](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 1

Project-wide support

## ofSupport

MathlibAnnex.CompactC1VectorField.ofSupport

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CompactC1VectorField](#)

### USED BY IN THIS PROJECT

[carrier\\_ofSupport](#), [ofSupport\\_apply](#), [translatedBumpField](#)

[Read exact source](#)

LEVEL 1

R07

## K\_nonneg

MathlibAnnex.DeterminantFrame.NearMaxInverseBound.boundConstant\_nonneg

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[NearMaxInverseBound](#)

### USED BY IN THIS PROJECT

[exists\\_net\\_preimage\\_close](#)

[Read exact source](#)

LEVEL 1

R07

## frameCoordinates

MathlibAnnex.DeterminantFrame.frameCoordinates

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#)

### USED BY IN THIS PROJECT

[frameCoordinates\\_eq\\_mulVec](#),  
[frameCoordinates\\_norm\\_le\\_model](#)

[Read exact source](#)

LEVEL 1

R07

## frameMap

MathlibAnnex.DeterminantFrame.frameMap

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#)

### USED BY IN THIS PROJECT

[frameMap\\_injective\\_of\\_det\\_ne\\_zero](#)

[Read exact source](#)

LEVEL 1

R07

## frameMatrix

MathlibAnnex.DeterminantFrame.frameMatrix

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#)

### USED BY IN THIS PROJECT

[frameCoordinates\\_eq\\_mulVec](#), [frameDeterminant](#),  
[frameMatrix\\_coordinateFrame](#)

▼ Show 1 more

[frameMatrix\\_replaceRow](#)

[Read exact source](#)

LEVEL 1

R07

## functional\_apply\_eq\_sum

MathlibAnnex.DeterminantFrame.functional\_apply\_eq\_sum

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateEquiv](#), [functionalCoordinates](#)

### USED BY IN THIS PROJECT

[cramerReplacement](#)

[Read exact source](#)

LEVEL 1

R07

## inverseCoordinateMap

MathlibAnnex.DeterminantFrame.inverseCoordinateMap

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateEquiv](#)

### USED BY IN THIS PROJECT

[inverseBoundConstant](#)

[Read exact source](#)

LEVEL 1

R07

## unitRowSet\_isCompact

MathlibAnnex.DeterminantFrame.isCompact\_unitRowSet

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[unitRowSet](#)

### USED BY IN THIS PROJECT

[unitFrameSet\\_isCompact](#)

[Read exact source](#)

LEVEL 1

R07

## mem\_unitRowSet

MathlibAnnex.DeterminantFrame.mem\_unitRowSet

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[unitRowSet](#)

### USED BY IN THIS PROJECT

[unitFrameSet\\_isCompact](#), [mem\\_unitFrameSet](#),  
[replaceRow\\_mem\\_unitFrameSet](#)

[Read exact source](#)

LEVEL 1

R07

## rawCoordinateRow

MathlibAnnex.DeterminantFrame.rawCoordinateRow

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateEquiv](#)

### USED BY IN THIS PROJECT

[coordinateRowSize](#)

[Read exact source](#)

LEVEL 1

R07

## replaceRow

MathlibAnnex.DeterminantFrame.replaceRow

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#)

### USED BY IN THIS PROJECT

[frameMatrix\\_replaceRow](#), [replaceRow\\_mem\\_unitFrameSet](#), [replacementDeterminant](#)

[Read exact source](#)

LEVEL 1

R07

## unitFrameSet

MathlibAnnex.DeterminantFrame.unitFrameSet

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#), [unitRowSet](#)

### USED BY IN THIS PROJECT

[unitFrameSet\\_isCompact](#), [mem\\_unitFrameSet](#), [replaceRow\\_mem\\_unitFrameSet](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## IsContraction

MathlibAnnex.EquivalentSeminorm.IsContraction

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[contractionSet](#), [zero\\_isContraction](#), [PluckerGeneratorGood](#)

▼ Show 2 more

[finMapConfiguration\\_mem](#), [generators](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

Project-wide support

## LinearIsometryEquiv

MathlibAnnex.EquivalentSeminorm.LinearIsometryEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[exists\\_linearIsometryEquiv\\_of\\_pluckerBodies\\_eq](#),  
[transportLinearIsometryEquiv](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## Space

MathlibAnnex.EquivalentSeminorm.Space

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[dist\\_space\\_eq](#), [norm\\_space\\_eq](#), [ofReference\\_toReference](#)

#### ▼ Show 11 more

[sphere\\_apply](#), [toReference\\_ofReference](#), [toReference\\_sub](#),  
[toReference\\_sub\\_ofReference](#), [toReference\\_zero](#),  
[CoefficientDetectsUnit](#),  
[absoluteMaximizer\\_satellite\\_norms\\_preimage](#),  
[frameCoordinates\\_mem\\_coefficientAnnulus](#),  
[modelDist\\_framePreimage\\_le](#), [satelliteBudget](#),  
[radialExtension\\_image\\_ball](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## closedUnitBall

MathlibAnnex.EquivalentSeminorm.closedUnitBall

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[closedUnitBallVolume](#), [mem\\_closedUnitBall](#),  
[zero\\_mem\\_closedUnitBall](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## eq\_zero\_of\_apply\_eq\_zero

MathlibAnnex.EquivalentSeminorm.eq\_zero\_of\_apply\_eq\_zero

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[dist\\_space\\_eq](#), [sphere\\_apply](#), [lower\\_of\\_ne\\_zero](#)

#### ▼ Show 6 more

[CoefficientDetectsUnit](#),  
[absoluteMaximizer\\_satellite\\_norms\\_preimage](#),  
[frameCoordinates\\_mem\\_coefficientAnnulus](#),  
[modelDist\\_framePreimage\\_le](#), [satelliteBudget](#),  
[radialExtension\\_image\\_ball](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## lipschitzWith\_upper\_of\_model\_bound

MathlibAnnex.EquivalentSeminorm.lipschitzWith\_upper\_of\_model\_bound

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 1

Project-wide support

## ofContinuousLinearEquiv

MathlibAnnex.EquivalentSeminorm.ofContinuousLinearEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[ofContinuousLinearEquiv\\_p\\_apply](#), [unitSphereEquiv](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## unitSphere

MathlibAnnex.EquivalentSeminorm.unitSphere

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[mem\\_unitSphere](#),  
[exists\\_extension\\_with\\_derivativeAverage\\_mem](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## ae\_norm\_apply\_le\_seminorm\_of\_lipschitz

MathlibAnnex.FDeriv.ae\_norm\_apply\_le\_seminorm\_of\_lipschitz

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[norm\\_apply\\_le\\_seminorm\\_of\\_lipschitz](#)

### USED BY IN THIS PROJECT

[derivativeGenerator\\_integrable\\_of\\_seminormLipschitz](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## AlmostIsometryMatch

MathlibAnnex.FiniteRecovery.AlmostIsometryMatch

Mathematical structure

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[exists\\_almostIsometryMatch\\_at\\_satelliteDimension](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## finMapBaseRow

MathlibAnnex.FiniteSup.Bridge.finMapBaseRow

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[satelliteAmbientDim](#)

### USED BY IN THIS PROJECT

[finMapConfiguration](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## finMapSatelliteRow

MathlibAnnex.FiniteSup.Bridge.finMapSatelliteRow

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[satelliteAmbientDim](#)

### USED BY IN THIS PROJECT

[finMapConfiguration](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## FinMapAlmostIsometric

MathlibAnnex.FiniteSup.FinMapAlmostIsometric

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[PluckerGeneratorGood](#), [lower\\_of\\_ne\\_zero](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

R03

## ofOrderEmbedding

MathlibAnnex.Matrix.MaximalMinorIndex.ofOrderEmbedding

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[MaximalMinorIndex](#)

### USED BY IN THIS PROJECT

[orderedRows\\_ofOrderEmbedding](#),  
[satellitePluckerCoefficients](#)

[Read exact source](#)

LEVEL 1

R03

## orderedRows

MathlibAnnex.Matrix.MaximalMinorIndex.orderedRows

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[MaximalMinorIndex](#)

### USED BY IN THIS PROJECT

[selectedOutputLinearMap](#),  
[orderedRows\\_ofOrderEmbedding](#), [maximalSubmatrix](#)

[Read exact source](#)

LEVEL 1

R09

## OrientedMaximalMinorsProportional

MathlibAnnex.Matrix.OrientedMaximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[orientedMaximalMinor](#)

### USED BY IN THIS PROJECT

[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#),  
[rowCoordinates\\_eq\\_of\\_orientedMaximalMinorsProportional](#),  
[scale\\_eq\\_one\\_of\\_orientedMaximalMinorsProportional\\_zero](#)

[Read exact source](#)

LEVEL 1

R03

## abs\_det\_le\_prod\_rowL1Norm

MathlibAnnex.Matrix.abs\_det\_le\_prod\_rowL1Norm

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[rowL1Norm](#)

### USED BY IN THIS PROJECT

[abs\\_det\\_sub\\_le\\_max](#)

[Read exact source](#)

**LEVEL 1**

R09

**chartFactor\_unique**

MathlibAnnex.Matrix.chartFactor\_unique

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[chartFactor](#), [orientedMaximalMinor](#)**USED BY IN THIS PROJECT**[existsUnique\\_factor\\_of\\_orientedMaximalMinorsProportional](#)[Read exact source](#)**LEVEL 1**

R09

**orientedMaximalMinor\_eq\_zero\_of\_not\_injective**

MathlibAnnex.Matrix.orientedMaximalMinor\_eq\_zero\_of\_not\_injective

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[orientedMaximalMinor](#)**USED BY IN THIS PROJECT**[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#)[Read exact source](#)**LEVEL 1**

R09

**orientedMaximalMinor\_permute**

MathlibAnnex.Matrix.orientedMaximalMinor\_permute

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[orientedMaximalMinor](#)**USED BY IN THIS PROJECT**[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#)[Read exact source](#)**LEVEL 1**

R09

**selectedSubmatrix\_mul\_chartFactor**

MathlibAnnex.Matrix.selectedSubmatrix\_mul\_chartFactor

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[chartFactor](#), [orientedMaximalMinor](#)**USED BY IN THIS PROJECT**[det\\_chartFactor](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**standardMollifier\_continuous**

MathlibAnnex.Mollification.continuous\_standardMollifier

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[standardMollifier](#)**USED BY IN THIS PROJECT**[mollify\\_add](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**standardMollifier\_hasCompactSupport**

MathlibAnnex.Mollification.hasCompactSupport\_standardMollifier

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[standardMollifier](#)**USED BY IN THIS PROJECT**[mollify\\_add](#)[Read exact source](#)

LEVEL 1

Project-wide support

## mollify

MathlibAnnex.Mollification.mollify

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[standardMollifier](#)

### USED BY IN THIS PROJECT

[mollify\\_contDiff](#),  
[eventually\\_tsupport\\_mollify\\_subset\\_compact](#), [mollify\\_add](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

R02

## le\_maximizer

MathlibAnnex.NonemptyCompacts.le\_maximizer

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizer](#)

### USED BY IN THIS PROJECT

[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#), [convexHull\\_maxSlice\\_eq\\_supportFace](#),  
[le\\_maxValue\\_of\\_mem\\_convexHull](#)

[Read exact source](#)

LEVEL 1

R02

## maxValue

MathlibAnnex.NonemptyCompacts.maxValue

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizer](#)

### USED BY IN THIS PROJECT

[maxSlice](#), [le\\_maxValue\\_of\\_mem\\_convexHull](#), [supportFace](#)

[Read exact source](#)

LEVEL 1

R02

## maximizer\_mem

MathlibAnnex.NonemptyCompacts.maximizer\_mem

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizer](#)

### USED BY IN THIS PROJECT

[maximizer\\_mem\\_maxSlice](#),  
[maxValue\\_eq\\_of\\_convexHull\\_eq](#)

[Read exact source](#)

LEVEL 1

R08

## map\_le

MathlibAnnex.NormBall.map\_le

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[map\\_le](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 1

R04

## basisRow\_apply

MathlibAnnex.Piola.basisRow\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[divergence\\_cofactorRowField\\_eq\\_zero](#)

[Read exact source](#)

LEVEL 1

R04

## cofactorRow

MathlibAnnex.Piola.cofactorRow

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[cofactorRowField](#),  
[det\\_updateRow\\_eq\\_sum\\_mul\\_cofactorRow](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

R04

## coordinateDivergence\_eq\_zero\_of\_not\_mem\_tsupport

MathlibAnnex.Piola.coordinateDivergence\_eq\_zero\_of\_not\_mem\_tsupport

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateDivergence](#)

### USED BY IN THIS PROJECT

[integral\\_coordinateDivergence\\_eq\\_zero\\_of\\_contDiff\\_hasCompactSupport](#)

[Read exact source](#)

LEVEL 1

R04

## hessianCoordinate

MathlibAnnex.Piola.hessianCoordinate

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[hessianCoordinate\\_comm](#)

[Read exact source](#)

LEVEL 1

R04

## outputHybrid\_empty

MathlibAnnex.Piola.outputHybrid\_empty

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[outputHybrid](#)

### USED BY IN THIS PROJECT

[integral\\_det\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)

[Read exact source](#)

LEVEL 1

R04

## outputHybrid\_univ

MathlibAnnex.Piola.outputHybrid\_univ

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[outputHybrid](#)

### USED BY IN THIS PROJECT

[integral\\_det\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)

[Read exact source](#)

LEVEL 1

R04

## singleOutputPerturb

MathlibAnnex.Piola.singleOutputPerturb

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[divergence\\_componentFlux\\_eq\\_det\\_sub\\_outputHybrid\\_insert\\_eq\\_singleOutputPerturb](#)

[Read exact source](#)

LEVEL 1

R04

## sum\_smul\_basisRow

MathlibAnnex.Piola.sum\_smul\_basisRow

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[det\\_updateRow\\_eq\\_sum\\_mul\\_cofactorRow](#)

[Read exact source](#)

LEVEL 1

R04

## twoRowReplacement

MathlibAnnex.Piola.twoRowReplacement

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[basisRow](#)

### USED BY IN THIS PROJECT

[det\\_twoRowReplacement\\_swap](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## LimitCertificate

MathlibAnnex.PluckerRecovery.LimitCertificate

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[measure\\_image\\_unitBall\\_eq](#),  
[exists\\_limitCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## LinearCertificate

MathlibAnnex.PluckerRecovery.LinearCertificate

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[modelNorm\\_le\\_two](#),  
[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## SatelliteConfiguration

MathlibAnnex.Satellite.SatelliteConfiguration

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#), [SatelliteRows](#)

### USED BY IN THIS PROJECT

[finMapConfiguration](#), [configurationMap](#),  
[configurationPolynomial](#)

▼ Show 1 more

[satelliteConfigurationSet](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## abs\_eval\_lower\_of\_norms\_nearby

MathlibAnnex.Satellite.abs\_eval\_lower\_of\_norms\_nearby

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#)

### USED BY IN THIS PROJECT

[absoluteMaximizer\\_satellites\\_almost\\_norm](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

Project-wide support

## satelliteRowsSet

MathlibAnnex.Satellite.satelliteRowsSet

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[EquivalentSeminorm](#), [SatelliteRows](#)

### USED BY IN THIS PROJECT

[satelliteConfigurationSet](#)

[Read exact source](#)

LEVEL 1

R08

## map\_eq

MathlibAnnex.SeminormBall.map\_eq

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[map\\_le](#)

### USED BY IN THIS PROJECT

[exists\\_linearIsometryEquiv\\_of\\_pluckerBodies\\_eq](#), [map\\_eq](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

R08

## measure\_lt

MathlibAnnex.SeminormBall.measure\_lt

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[interior\\_sdiff\\_nonempty](#)

### USED BY IN THIS PROJECT

[measure\\_lt](#), [image\\_unitBall\\_eq](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

Project-wide support

## coe\_normalizeToSphere

MathlibAnnex.Sphere.coe\_normalizeToSphere

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[normalizeToSphere](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 1

Project-wide support

## normalizeToSphere\_pos\_smul

MathlibAnnex.Sphere.normalizeToSphere\_pos\_smul

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[normalizeToSphere](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 1

Project-wide support

## normalizeToSphere\_unit

MathlibAnnex.Sphere.normalizeToSphere\_unit

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[normalizeToSphere](#)

### USED BY IN THIS PROJECT

[radialExtension\\_on\\_sphere](#)

[Read exact source](#)

**LEVEL 1**

R01

**radialExtension**`MathlibAnnex.Sphere.radialExtension`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[normalizeToSphere](#)**USED BY IN THIS PROJECT**[radialExtension\\_of\\_ne\\_zero](#), [radialExtension\\_zero](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 1**

Project-wide support

**carrier**`MathlibAnnex.WeakGradient.LocalBallData.carrier`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LocalBallData](#)**USED BY IN THIS PROJECT**[carrier\\_subset](#), [inner\\_subset\\_carrier](#), [carrier\\_isCompact](#)**▼ Show 2 more**[carrier\\_measurable](#), [middle\\_subset\\_carrier](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**inner**`MathlibAnnex.WeakGradient.LocalBallData.inner`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LocalBallData](#)**USED BY IN THIS PROJECT**[center\\_mem\\_inner](#), [inner\\_subset\\_carrier](#), [inner\\_isOpen](#)**▼ Show 1 more**[translated\\_closedBall\\_subset\\_middle](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**middle**`MathlibAnnex.WeakGradient.LocalBallData.middle`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LocalBallData](#)**USED BY IN THIS PROJECT**[middle\\_subset\\_carrier](#),  
[translated\\_closedBall\\_subset\\_middle](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**compactLocalization\_of\_mem**`MathlibAnnex.WeakGradient.compactLocalization_of_mem`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[compactLocalization](#)**USED BY IN THIS PROJECT**[compactLocalization\\_eq\\_ae\\_inner](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**compactLocalization\_of\_not\_mem**`MathlibAnnex.WeakGradient.compactLocalization_of_not_mem`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[compactLocalization](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

**LEVEL 1**

Project-wide support

**localMollification**`MathlibAnnex.WeakGradient.localMollification`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump](#)**USED BY IN THIS PROJECT**[localMollification\\_tendsto\\_ae](#), [localMollification\\_contDiff](#),  
[localMollification\\_fderiv\\_apply](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**exists\_localBallData**`MathlibAnnex.WeakGradient.nonempty_localBallData`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LocalBallData](#)**USED BY IN THIS PROJECT**[exists\\_localAECConstantAt](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**normalizedShrinkingBump**`MathlibAnnex.WeakGradient.normalizedShrinkingBump`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump](#)**USED BY IN THIS PROJECT**[localMollification\\_fderiv\\_apply](#), [translatedBumpField](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**shrinkingBump\_rIn**`MathlibAnnex.WeakGradient.shrinkingBump_rIn`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump](#)**USED BY IN THIS PROJECT**[shrinkingBump\\_ratio](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**shrinkingBump\_rOut**`MathlibAnnex.WeakGradient.shrinkingBump_rOut`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump](#)**USED BY IN THIS PROJECT**[shrinkingBump\\_ratio](#)[Read exact source](#)**LEVEL 1**

Project-wide support

**shrinkingBump\_rOut\_le\_half**`MathlibAnnex.WeakGradient.shrinkingBump_rOut_le_half`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump](#)**USED BY IN THIS PROJECT**[translatedBumpField\\_carrier\\_subset](#)[Read exact source](#)

LEVEL 1

Project-wide support

## shrinkingBump\_rOut\_tendsto\_zero

MathlibAnnex.WeakGradient.tendsto\_shrinkingBump\_rOut\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[shrinkingBump](#)

### USED BY IN THIS PROJECT

[localMollification\\_tendsto\\_ae](#)

[Read exact source](#)

LEVEL 1

R05

## aeConstantOn\_of\_countableCover

MathlibAnnex.aeConstantOn\_of\_countableCover

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[AEConstantOn](#)

### USED BY IN THIS PROJECT

[aeConstantOn\\_of\\_everywhere\\_local](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

R05

## aeConstants\_eq\_of\_open\_overlap

MathlibAnnex.aeConstants\_eq\_of\_open\_overlap

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[AEConstantOn](#)

### USED BY IN THIS PROJECT

[constantRegion\\_compl\\_isOpen](#)

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LEVEL 1

R02

## allCoordinates

MathlibAnnex.allCoordinates

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinate](#)

### USED BY IN THIS PROJECT

[coordinate\\_mem\\_allCoordinates](#)

[Read exact source](#)

LEVEL 1

R05

## ConstantRegion

MathlibAnnex.constantRegion

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[AEConstantOn](#)

### USED BY IN THIS PROJECT

[constantRegion\\_isOpen](#), [constantRegion\\_compl\\_isOpen](#)

[Read exact source](#)

LEVEL 1

Project-wide support

## divergence\_pi

MathlibAnnex.divergence\_pi

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[divergence](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#)

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**LEVEL 1**

Project-wide support

**divergence\_sub\_smul**

MathlibAnnex.divergence\_sub\_smul

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[divergence](#), [trace\\_smulRight](#)**USED BY IN THIS PROJECT**[divergence\\_translatedBumpField](#)[Read exact source](#)**Level 2** 77 declarations**LEVEL 2**

Project-wide support

**abs\_domainJacobianSign**

MathlibAnnex.BilipschitzOrientation.abs\_domainJacobianSign

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[domainJacobianSign](#)**USED BY IN THIS PROJECT**[abs\\_targetJacobianSign](#), [signed\\_area\\_transfer](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**fderiv\_injective**

MathlibAnnex.BilipschitzOrientation.fderiv\_injective

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[fderiv\\_lower\\_bound](#)**USED BY IN THIS PROJECT**[det\\_fderiv\\_ne\\_zero](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**targetJacobianSign**

MathlibAnnex.BilipschitzOrientation.targetJacobianSign

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[domainJacobianSign](#)**USED BY IN THIS PROJECT**[abs\\_targetJacobianSign](#), [signed\\_area\\_transfer](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**carrier\_ofSupport**

MathlibAnnex.CompactC1VectorField.carrier\_ofSupport

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#), [ofSupport](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

LEVEL 2

Project-wide support

## continuous

MathlibAnnex.CompactC1VectorField.continuous

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[contDiff](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 2

Project-wide support

## carrier\_compact

MathlibAnnex.CompactC1VectorField.isCompact\_carrier

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[carrier](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## ofSupport\_apply

MathlibAnnex.CompactC1VectorField.ofSupport\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ofSupport](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 2

Project-wide support

## support\_subset

MathlibAnnex.CompactC1VectorField.support\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[carrier](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## tsupport\_subset

MathlibAnnex.CompactC1VectorField.tsupport\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[carrier](#)

### USED BY IN THIS PROJECT

[fderiv\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#)

[Read exact source](#)

LEVEL 2

R03

## abs\_det\_sub\_le\_max

MathlibAnnex.ContinuousLinearMap.abs\_det\_sub\_le\_max

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_det\\_le\\_prod\\_rowL1Norm](#)

### USED BY IN THIS PROJECT

[abs\\_det\\_sub\\_le](#)

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LEVEL 2

Project-wide support

## selectedOutputLinearMap

MathlibAnnex.ContinuousLinearMap.selectedOutputLinearMap

Mathematical construction

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[orderedRows](#)

USED BY IN THIS PROJECT  
[norm\\_selectedOutputLinearMap\\_le](#)

[Read exact source](#)

LEVEL 2

R07

## coordinateRowSize

MathlibAnnex.DeterminantFrame.coordinateRowSize

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[rawCoordinateRow](#)

USED BY IN THIS PROJECT  
[coordinateRowSize\\_nonneg](#), [coordinateScale](#),  
[norm\\_rawCoordinateRow\\_le\\_size](#)

[Read exact source](#)

LEVEL 2

R07

## frameCoordinates\_eq\_mulVec

MathlibAnnex.DeterminantFrame.frameCoordinates\_eq\_mulVec

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[coordinateEquiv](#), [frameCoordinates](#), [frameMatrix](#)

USED BY IN THIS PROJECT  
[frameMap\\_injective\\_of\\_det\\_ne\\_zero](#),  
[inverse\\_mulVec\\_frameCoordinates](#)

[Read exact source](#)

LEVEL 2

R07

## frameDeterminant

MathlibAnnex.DeterminantFrame.frameDeterminant

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[frameMatrix](#)

USED BY IN THIS PROJECT  
[frameDeterminant\\_continuous](#),  
[frameDeterminant\\_coordinateFrame](#),  
[frameMap\\_injective\\_of\\_det\\_ne\\_zero](#)

▼ Show 1 more

[replacementDeterminant](#)

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LEVEL 2

R07

## frameMatrix\_replaceRow

MathlibAnnex.DeterminantFrame.frameMatrix\_replaceRow

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[frameMatrix](#), [functionalCoordinates](#), [replaceRow](#)

USED BY IN THIS PROJECT  
[replacementDeterminant\\_eq\\_cramerTranspose](#),  
[pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#),  
[absoluteMaximizer\\_base\\_nearMax](#)

[Read exact source](#)

LEVEL 2

R07

## unitFrameSet\_isCompact

MathlibAnnex.DeterminantFrame.isCompact\_unitFrameSet

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[unitRowSet\\_isCompact](#), [mem\\_unitRowSet](#), [unitFrameSet](#)

USED BY IN THIS PROJECT  
[exists\\_maximizingFrame](#)

[Read exact source](#)

**LEVEL 2**

R07

**mem\_unitFrameSet**`MathlibAnnex.DeterminantFrame.mem_unitFrameSet`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[mem\\_unitRowSet](#), [unitFrameSet](#)**USED BY IN THIS PROJECT**[coordinateFrame\\_mem](#), [unitFrameSet\\_nonempty](#)[Read exact source](#)**LEVEL 2**

R07

**replaceRow\_mem\_unitFrameSet**`MathlibAnnex.DeterminantFrame.replaceRow_mem_unitFrameSet`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[mem\\_unitRowSet](#), [replaceRow](#), [unitFrameSet](#)**USED BY IN THIS PROJECT**[abs\\_replacementDeterminant\\_le\\_maximum](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**closedUnitBallVolume**`MathlibAnnex.EquivalentSeminorm.closedUnitBallVolume`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[closedUnitBall](#)**USED BY IN THIS PROJECT**[closedUnitBallVolume\\_pos](#),  
[ballVolumeScaledMaximalMinors](#),  
[measure\\_image\\_unitBall\\_eq](#)**▼ Show 1 more**[radialExtension\\_integral\\_abs\\_det](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**contractionSet**`MathlibAnnex.EquivalentSeminorm.contractionSet`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[IsContraction](#)**USED BY IN THIS PROJECT**[mem\\_contractionSet](#), [generators\\_eq\\_image\\_union](#)[Read exact source](#)

**LEVEL 2**

Project-wide support

**dist\_space\_eq**

MathlibAnnex.EquivalentSeminorm.dist\_space\_eq

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#), [eq\\_zero\\_of\\_apply\\_eq\\_zero](#)**USED BY IN THIS PROJECT**[spherIsometryEquiv](#),  
[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#),  
[exists\\_extension\\_with\\_derivativeAverage\\_mem](#)

▼ Show 1 more

[radialExtension\\_integral\\_abs\\_det](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**zero\_isContraction**

MathlibAnnex.EquivalentSeminorm.isContraction\_zero

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[IsContraction](#)**USED BY IN THIS PROJECT**[generators\\_nonempty](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**mem\_closedUnitBall**

MathlibAnnex.EquivalentSeminorm.mem\_closedUnitBall

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[closedUnitBall](#)**USED BY IN THIS PROJECT**[closedUnitBall\\_isBounded](#),  
[zero\\_mem\\_interior\\_closedUnitBall](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**mem\_unitSphere**

MathlibAnnex.EquivalentSeminorm.mem\_unitSphere

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[unitSphere](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**norm\_space\_eq**

MathlibAnnex.EquivalentSeminorm.norm\_space\_eq

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**ofContinuousLinearEquiv\_p\_apply**

MathlibAnnex.EquivalentSeminorm.ofContinuousLinearEquiv\_p\_apply

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ofContinuousLinearEquiv](#)**USED BY IN THIS PROJECT**[transportLinearIsometryEquiv](#)[Read exact source](#)

**LEVEL 2**

Project-wide support

**ofReference\_toReference**`MathlibAnnex.EquivalentSeminorm.ofReference_toReference`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)**LEVEL 2**

Project-wide support

**sphere\_apply**`MathlibAnnex.EquivalentSeminorm.sphere_apply`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#), [eq\\_zero\\_of\\_apply\\_eq\\_zero](#)**USED BY IN THIS PROJECT**[unitSphereEquiv](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**toReference\_ofReference**`MathlibAnnex.EquivalentSeminorm.toReference_ofReference`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**toReference\_sub**`MathlibAnnex.EquivalentSeminorm.toReference_sub`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)**LEVEL 2**

Project-wide support

**toReference\_sub\_ofReference**`MathlibAnnex.EquivalentSeminorm.toReference_sub_ofReference`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**toReference\_zero**`MathlibAnnex.EquivalentSeminorm.toReference_zero`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[Space](#)**USED BY IN THIS PROJECT**

None in this Project

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LEVEL 2

Project-wide support

## zero\_mem\_closedUnitBall

MathlibAnnex.EquivalentSeminorm.zero\_mem\_closedUnitBall

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBall](#)

### USED BY IN THIS PROJECT

[closedUnitBall\\_nonempty](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## finMapConfiguration

MathlibAnnex.FiniteSup.Bridge.finMapConfiguration

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[finMapBaseRow](#), [finMapSatelliteRow](#), [SatelliteConfiguration](#)

### USED BY IN THIS PROJECT

[configurationFinMap\\_finMapConfiguration](#),  
[finMapConfiguration\\_base\\_apply](#),  
[finMapConfiguration\\_configurationFinMap](#)

▼ Show 2 more

[finMapConfiguration\\_mem](#),  
[finMapConfiguration\\_satellite\\_apply](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## lower\_of\_ne\_zero

MathlibAnnex.FiniteSup.FinMapAlmostIsometric.lower\_of\_ne\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[eq\\_zero\\_of\\_apply\\_eq\\_zero](#), [FinMapAlmostIsometric](#)

### USED BY IN THIS PROJECT

[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## configurationMap

MathlibAnnex.FiniteSup.configurationMap

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[SatelliteConfiguration](#)

### USED BY IN THIS PROJECT

[configurationFinMap](#),  
[configurationMap\\_norm\\_gt\\_of\\_satellite](#),  
[configurationMap\\_norm\\_le\\_model](#)

[Read exact source](#)

LEVEL 2

R03

## orderedRows\_ofOrderEmbedding

MathlibAnnex.Matrix.MaximalMinorIndex.orderedRows\_ofOrderEmbedding

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ofOrderEmbedding](#), [orderedRows](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#), [maximalSubmatrix\\_ofOrderEmbedding](#),  
[radialExtension\\_integral\\_abs\\_det](#)

[Read exact source](#)

LEVEL 2

R09

## det\_chartFactor

MathlibAnnex.Matrix.det\_chartFactor

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedSubmatrix\\_mul\\_chartFactor](#)

### USED BY IN THIS PROJECT

[existsUnique\\_factor\\_of\\_orientedMaximalMinorsProportional](#),  
[isUnit\\_chartFactor\\_of\\_scale\\_ne\\_zero](#)

[Read exact source](#)

LEVEL 2

R03

## maximalSubmatrix

MathlibAnnex.Matrix.maximalSubmatrix

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[orderedRows](#)

### USED BY IN THIS PROJECT

[maximalMinor](#), [maximalSubmatrix\\_mul](#),  
[maximalSubmatrix\\_mulVec](#)

#### ▼ Show 2 more

[maximalSubmatrix\\_ofOrderEmbedding](#),  
[maximalSubmatrix\\_zero](#)

[Read exact source](#)

LEVEL 2

R09

## rowCoordinates\_eq\_of\_orientedMaximalMinorsProportional

MathlibAnnex.Matrix.rowCoordinates\_eq\_of\_orientedMaximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[OrientedMaximalMinorsProportional](#),  
[det\\_smul\\_vecMul\\_nonsingInv\\_eq\\_updateRowDet](#),  
[submatrix\\_update\\_rowTuple](#)

### USED BY IN THIS PROJECT

[mul\\_chartFactor\\_eq\\_of\\_orientedMaximalMinorsProportional](#)

[Read exact source](#)

LEVEL 2

R09

## scale\_eq\_one\_of\_orientedMaximalMinorsProportional\_zero

MathlibAnnex.Matrix.scale\_eq\_one\_of\_orientedMaximalMinorsProportional\_zero

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[OrientedMaximalMinorsProportional](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 2

Project-wide support

## mollify\_contDiff

MathlibAnnex.Mollification.contDiff\_mollify

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mollify](#)

### USED BY IN THIS PROJECT

[eventually\\_mollify\\_fderiv\\_memLpOn\\_compact](#),  
[maximalMinorIntegrand\\_mollify\\_continuous](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## eventually\_tsupport\_mollify\_subset\_compact

MathlibAnnex.Mollification.eventually\_tsupport\_mollify\_subset\_compact

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mollify](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## mollify\_add

MathlibAnnex.Mollification.mollify\_add

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[standardMollifier\\_continuous](#),  
[standardMollifier\\_hasCompactSupport](#), [mollify](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

**LEVEL 2**

R02

**maxSlice**`MathlibAnnex.NonemptyCompacts.maxSlice`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[maxValue](#)**USED BY IN THIS PROJECT**[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#), [maxSlice\\_isCompact](#), [maximizer\\_mem\\_maxSlice](#)

▼ Show 1 more

[convexHull\\_maxSlice\\_eq\\_supportFace](#)[Read exact source](#)**LEVEL 2**

R08

**map\_eq**`MathlibAnnex.NormBall.map_eq`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[map\\_eq](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)**LEVEL 2**

R08

**measure\_lt**`MathlibAnnex.NormBall.measure_lt`

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[measure\\_lt](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)**LEVEL 2**

R04

**cofactorRowField**`MathlibAnnex.Piola.cofactorRowField`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[cofactorRow](#)**USED BY IN THIS PROJECT**[componentFlux](#), [piola\\_divergence\\_eq\\_zero\\_of\\_expansion](#)[Read exact source](#)**LEVEL 2**

R04

**det\_twoRowReplacement\_swap**`MathlibAnnex.Piola.det_twoRowReplacement_swap`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[twoRowReplacement](#)**USED BY IN THIS PROJECT**[sum\\_hessian\\_twoRowReplacement\\_eq\\_zero](#)[Read exact source](#)**LEVEL 2**

R04

**det\_updateRow\_eq\_sum\_mul\_cofactorRow**`MathlibAnnex.Piola.det_updateRow_eq_sum_mul_cofactorRow`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[cofactorRow](#), [det\\_updateRow\\_finset\\_sum](#), [sum\\_smul\\_basisRow](#)**USED BY IN THIS PROJECT**[divergence\\_componentFlux\\_eq\\_det\\_sub](#)[Read exact source](#)

**LEVEL 2**

R04

**hessianCoordinate\_comm**

MathlibAnnex.Piola.hessianCoordinate\_comm

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[hessianCoordinate](#)**USED BY IN THIS PROJECT**[sum\\_hessianCoordinate\\_twoRowReplacement\\_eq\\_zero](#)[Read exact source](#)**LEVEL 2**

R04

**integral\_coordinateDivergence\_eq\_zero\_of\_contDiff\_hasCompactSupport**

MathlibAnnex.Piola.integral\_coordinateDivergence\_eq\_zero\_of\_contDiff\_hasCompactSupport

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinateDivergence\\_eq\\_zero\\_of\\_not\\_mem\\_tsupport,](#)  
[exists\\_open\\_box\\_containing\\_compact](#)**USED BY IN THIS PROJECT**[integral\\_det\\_singleOutputPerturb\\_sub\\_eq\\_zero](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 2**

R04

**outputHybrid\_insert\_eq\_singleOutputPerturb**

MathlibAnnex.Piola.outputHybrid\_insert\_eq\_singleOutputPerturb

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[outputHybrid,](#) [singleOutputPerturb](#)**USED BY IN THIS PROJECT**[integral\\_det\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**modelNorm\_le\_two**MathlibAnnex.PluckerRecovery.LinearCertificate.s  
eminorm\_linearMap\_le\_two\_mul

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LinearCertificate](#)**USED BY IN THIS PROJECT**[referenceNorm\\_le](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**satellitePluckerCoefficients**

MathlibAnnex.PluckerSupport.satellitePluckerCoefficients

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ofOrderEmbedding](#)**USED BY IN THIS PROJECT**[orientedSatelliteSupport,](#)  
[pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**frameCoordinates\_norm\_le\_model**

MathlibAnnex.Satellite.frameCoordinates\_norm\_le\_model

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[frameCoordinates,](#) [EquivalentSeminorm](#)**USED BY IN THIS PROJECT**[frameCoordinates\\_mem\\_coefficientAnnulus](#)[Read exact source](#)

LEVEL 2

Project-wide support

## satelliteConfigurationSet

MathlibAnnex.Satellite.satelliteConfigurationSet

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[SatelliteConfiguration](#), [satelliteRowsSet](#)

### USED BY IN THIS PROJECT

[finMapConfiguration\\_mem](#),  
[configurationMap\\_norm\\_le\\_model](#),  
[absoluteMaximizer\\_base\\_nearMax](#)

#### ▼ Show 2 more

[absoluteMaximizer\\_satellite\\_norms\\_preimage](#),  
[maxSatelliteConfiguration](#)

[Read exact source](#)

LEVEL 2

R01

## radialExtension\_of\_ne\_zero

MathlibAnnex.Sphere.radialExtension\_of\_ne\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension](#)

### USED BY IN THIS PROJECT

[radialExtension\\_norm](#), [radialExtension\\_on\\_sphere](#)

[Read exact source](#)

LEVEL 2

R01

## radialExtension\_zero

MathlibAnnex.Sphere.radialExtension\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension](#)

### USED BY IN THIS PROJECT

[radialExtension\\_norm](#)

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LEVEL 2

Project-wide support

## WeakDivergenceZero

MathlibAnnex.WeakDivergenceZero

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[carrier](#), [divergence](#)

### USED BY IN THIS PROJECT

[targetJacobianSign\\_weakDivergenceZero](#),  
[weakIntegral\\_eq\\_localized\\_bumpIntegral](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 2

Project-wide support

## carrier\_subset

MathlibAnnex.WeakGradient.LocalBallData.carrier\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[carrier](#)

### USED BY IN THIS PROJECT

[middle\\_subset\\_open](#), [integrableOn\\_localBallCarrier](#)

[Read exact source](#)

LEVEL 2

Project-wide support

## center\_mem\_inner

MathlibAnnex.WeakGradient.LocalBallData.center\_mem\_inner

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[inner](#)

### USED BY IN THIS PROJECT

[inner\\_nonempty](#)

[Read exact source](#)

**LEVEL 2**

Project-wide support

**inner\_subset\_carrier**`MathlibAnnex.WeakGradient.LocalBallData.inner_subset_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#), [inner](#)**USED BY IN THIS PROJECT**[compactLocalization\\_eq\\_ae\\_inner](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**carrier\_isCompact**`MathlibAnnex.WeakGradient.LocalBallData.isCompact_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#)**USED BY IN THIS PROJECT**[integrableOn\\_localBallCarrier](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**inner\_isOpen**`MathlibAnnex.WeakGradient.LocalBallData.isOpen_inner`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[inner](#)**USED BY IN THIS PROJECT**[compactLocalization\\_eq\\_ae\\_inner](#),  
[exists\\_localMollification\\_constant](#),  
[localBall\\_inner\\_measure\\_pos](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**carrier\_measurable**`MathlibAnnex.WeakGradient.LocalBallData.measurableSet_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#)**USED BY IN THIS PROJECT**[compactLocalization\\_integrable](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**middle\_subset\_carrier**`MathlibAnnex.WeakGradient.LocalBallData.middle_subset_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#), [middle](#)**USED BY IN THIS PROJECT**[middle\\_subset\\_open](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**translated\_closedBall\_subset\_middle**`MathlibAnnex.WeakGradient.LocalBallData.translated_closedBall_subset_middle`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[inner](#), [middle](#)**USED BY IN THIS PROJECT**[translatedBumpField\\_carrier\\_subset](#)[Read exact source](#)

**LEVEL 2**

Project-wide support

**localMollification\_fderiv\_apply**`MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[localMollification](#), [normalizedShrinkingBump](#)**USED BY IN THIS PROJECT**[localMollification\\_fderiv\\_apply\\_eq\\_zero](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 2**

Project-wide support

**shrinkingBump\_ratio**`MathlibAnnex.WeakGradient.shrinkingBump_ratio`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[shrinkingBump\\_rIn](#), [shrinkingBump\\_rOut](#)**USED BY IN THIS PROJECT**[localMollification\\_tendsto\\_ae](#)[Read exact source](#)**LEVEL 2**

Project-wide support

**translatedBumpField**`MathlibAnnex.WeakGradient.translatedBumpField`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ofSupport](#), [normalizedShrinkingBump](#)**USED BY IN THIS PROJECT**[divergence\\_translatedBumpField](#),  
[translatedBumpField\\_carrier\\_subset](#)[Read exact source](#)**LEVEL 2**

R05

**aeConstantOn\_of\_everywhere\_local**`MathlibAnnex.aeConstantOn_of_everywhere_local`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[aeConstantOn\\_of\\_countableCover](#)**USED BY IN THIS PROJECT**[exists\\_aeConstantOn\\_of\\_local](#)[Read exact source](#)**LEVEL 2**

R02

**coordinate\_mem\_allCoordinates**`MathlibAnnex.coordinate_mem_allCoordinates`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[allCoordinates](#)**USED BY IN THIS PROJECT**[allCoordinates\\_separate](#)[Read exact source](#)**LEVEL 2**

R05

**constantRegion\_isOpen**`MathlibAnnex.isOpen_constantRegion`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ConstantRegion](#)**USED BY IN THIS PROJECT**[constantRegion\\_eq\\_of\\_preconnected](#)[Read exact source](#)

**LEVEL 2**

R05

**constantRegion\_compl\_isOpen**

MathlibAnnex.isOpen\_constantRegion\_compl

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[LocalAECongruentAt](#), [aeConstants\\_eq\\_of\\_open\\_overlap](#),  
[ConstantRegion](#)**USED BY IN THIS PROJECT**[constantRegion\\_eq\\_of\\_preconnected](#)[Read exact source](#)**LEVEL 2**

R02

**le\_maxValue\_of\_mem\_convexHull**

MathlibAnnex.le\_maxValue\_of\_mem\_convexHull

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[le\\_maximizer](#), [maxValue](#)**USED BY IN THIS PROJECT**[maxValue\\_eq\\_of\\_convexHull\\_eq](#)[Read exact source](#)**LEVEL 2**

R02

**supportFace**

MathlibAnnex.supportFace

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[maxValue](#)**USED BY IN THIS PROJECT**[convexHull\\_maxSlice\\_eq\\_supportFace](#)[Read exact source](#)**Level 3** 48 declarations**LEVEL 3**

Project-wide support

**abs\_targetJacobianSign**

MathlibAnnex.BilipschitzOrientation.abs\_targetJacobianSign

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_domainJacobianSign](#), [targetJacobianSign](#)**USED BY IN THIS PROJECT**[targetJacobianSign\\_locallyIntegrableOn](#),  
[targetJacobianSign\\_const\\_is\\_pm\\_one](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**det\_fderiv\_ne\_zero**

MathlibAnnex.BilipschitzOrientation.det\_fderiv\_ne\_zero

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[fderiv\\_injective](#)**USED BY IN THIS PROJECT**[det\\_eq\\_domainSign\\_mul\\_abs](#)[Read exact source](#)

**LEVEL 3**

Project-wide support

**fderiv\_eq\_zero\_of\_not\_mem\_carrier**`MathlibAnnex.CompactC1VectorField.fderiv_eq_zero_of_not_mem_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[tsupport\\_subset](#)**USED BY IN THIS PROJECT**[weak\\_piola](#), [divergence\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#), [weakIntegral\\_eq\\_localized\\_bumpIntegral](#)[Read exact source](#)**LEVEL 3**

R03

**abs\_det\_sub\_le**`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_det\\_sub\\_le\\_max](#)**USED BY IN THIS PROJECT**[norm\\_det\\_sub\\_le](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**norm\_selectedOutputLinearMap\_le**`MathlibAnnex.ContinuousLinearMap.norm_selectedOutputLinearMap_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[selectedOutputLinearMap](#)**USED BY IN THIS PROJECT**[selectedOutput](#)[Read exact source](#)**LEVEL 3**

R07

**frameDeterminant\_continuous**`MathlibAnnex.DeterminantFrame.continuous_frameDeterminant`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[frameDeterminant](#)**USED BY IN THIS PROJECT**[exists\\_maximizingFrame](#), [maxSatelliteConfiguration](#)[Read exact source](#)**LEVEL 3**

R07

**coordinateRowSize\_nonneg**`MathlibAnnex.DeterminantFrame.coordinateRowSize_nonneg`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinateRowSize](#)**USED BY IN THIS PROJECT**[coordinateScale\\_pos](#)[Read exact source](#)**LEVEL 3**

R07

**coordinateScale**`MathlibAnnex.DeterminantFrame.coordinateScale`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinateRowSize](#)**USED BY IN THIS PROJECT**[coordinateInverseFactor](#), [coordinateRow](#), [coordinateScale\\_pos](#)[Read exact source](#)

LEVEL 3

R07

## norm\_rawCoordinateRow\_le\_size

MathlibAnnex.DeterminantFrame.norm\_rawCoordinateRow\_le\_size

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateRowSize](#)

### USED BY IN THIS PROJECT

[coordinateFrame\\_mem](#)

[Read exact source](#)

LEVEL 3

R07

## replacementDeterminant

MathlibAnnex.DeterminantFrame.replacementDeterminant

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameDeterminant](#), [replaceRow](#)

### USED BY IN THIS PROJECT

[abs\\_replacementDeterminant\\_le\\_maximum](#),  
[replacementDeterminant\\_eq\\_cramerTranspose](#),  
[satellitePolynomial](#)

[Read exact source](#)

LEVEL 3

R07

## unitFrameSet\_nonempty

MathlibAnnex.DeterminantFrame.unitFrameSet\_nonempty

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mem\\_unitFrameSet](#)

### USED BY IN THIS PROJECT

[exists\\_maximizingFrame](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## closedUnitBall\_nonempty

MathlibAnnex.EquivalentSeminorm.closedUnitBall\_nonempty

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[zero\\_mem\\_closedUnitBall](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 3

Project-wide support

## closedUnitBall\_isBounded

MathlibAnnex.EquivalentSeminorm.isBounded\_closedUnitBall

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mem\\_closedUnitBall](#)

### USED BY IN THIS PROJECT

[closedUnitBall\\_isCompact](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## mem\_contractionSet

MathlibAnnex.EquivalentSeminorm.mem\_contractionSet

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[contractionSet](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 3

Project-wide support

## transportLinearIsometryEquiv

MathlibAnnex.EquivalentSeminorm.transportLinearIsometryEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[LinearIsometryEquiv](#), [ofContinuousLinearEquiv\\_p\\_apply](#)

### USED BY IN THIS PROJECT

[linearIsometryEquiv\\_of\\_coordinate\\_model](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## unitSphereEquiv

MathlibAnnex.EquivalentSeminorm.unitSphereEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ofContinuousLinearEquiv](#), [sphere\\_apply](#)

### USED BY IN THIS PROJECT

[sphereIsometryEquiv](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## zero\_mem\_interior\_closedUnitBall

MathlibAnnex.EquivalentSeminorm.zero\_mem\_interior\_closedUnitBall

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mem\\_closedUnitBall](#)

### USED BY IN THIS PROJECT

[closedUnitBallVolume\\_pos](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## configurationFinMap

MathlibAnnex.FiniteSup.Bridge.configurationFinMap

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[satelliteAmbientDim](#), [configurationMap](#)

### USED BY IN THIS PROJECT

[configurationFinMap\\_apply\\_base](#),  
[configurationFinMap\\_apply\\_satellite](#),  
[configurationFinMap\\_finMapConfiguration](#)

▼ Show 1 more

[configurationFinMap\\_norm](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## finMapConfiguration\_base\_apply

MathlibAnnex.FiniteSup.Bridge.finMapConfiguration\_base\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[finMapConfiguration](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 3

Project-wide support

## finMapConfiguration\_mem

MathlibAnnex.FiniteSup.Bridge.finMapConfiguration\_mem

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[IsContraction](#), [finMapConfiguration](#), [satelliteConfigurationSet](#)

### USED BY IN THIS PROJECT

[configuration\\_contraction\\_iff](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## finMapConfiguration\_satellite\_apply

MathlibAnnex.FiniteSup.Bridge.finMapConfiguration\_satellite\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[finMapConfiguration](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 3

Project-wide support

## configurationMap\_norm\_gt\_of\_satellite

MathlibAnnex.FiniteSup.configurationMap\_norm\_gt\_of\_satellite

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationMap](#)

### USED BY IN THIS PROJECT

[maxNetConfigurationMap\\_lower\\_of\\_ne\\_zero](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## configurationMap\_norm\_le\_model

MathlibAnnex.FiniteSup.configurationMap\_norm\_le\_model

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationMap](#), [satelliteConfigurationSet](#)

### USED BY IN THIS PROJECT

[configurationFinMap\\_norm\\_le\\_model](#)

[Read exact source](#)

LEVEL 3

R09

## isUnit\_chartFactor\_of\_scale\_ne\_zero

MathlibAnnex.Matrix.isUnit\_chartFactor\_of\_scale\_ne\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[det\\_chartFactor](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 3

R03

## maximalMinor

MathlibAnnex.Matrix.maximalMinor

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalSubmatrix](#)

### USED BY IN THIS PROJECT

[det\\_selectedSquare](#), [MaximalMinorsProportional](#), [maximalMinor\\_mul](#)

▼ Show 4 more

[maximalMinor\\_ofOrderEmbedding](#), [maximalMinor\\_zero\\_of\\_pos](#), [maximalMinors](#), [mulVec\\_injective\\_of\\_maximalMinor\\_ne\\_zero](#)

[Read exact source](#)

LEVEL 3

R03

## maximalSubmatrix\_mul

MathlibAnnex.Matrix.maximalSubmatrix\_mul

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalSubmatrix](#)

### USED BY IN THIS PROJECT

[maximalMinor\\_mul](#)

[Read exact source](#)

LEVEL 3

R03

## maximalSubmatrix\_mulVec

MathlibAnnex.Matrix.maximalSubmatrix\_mulVec

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalSubmatrix](#)

### USED BY IN THIS PROJECT

[mulVec\\_injective\\_of\\_maximalMinor\\_ne\\_zero](#)

[Read exact source](#)

LEVEL 3

R03

## maximalSubmatrix\_ofOrderEmbedding

MathlibAnnex.Matrix.maximalSubmatrix\_ofOrderEmbedding

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[orderedRows\\_ofOrderEmbedding](#), [maximalSubmatrix](#)

### USED BY IN THIS PROJECT

[maximalMinor\\_ofOrderEmbedding](#)

[Read exact source](#)

LEVEL 3

R03

## maximalSubmatrix\_zero

MathlibAnnex.Matrix.maximalSubmatrix\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalSubmatrix](#)

### USED BY IN THIS PROJECT

[maximalMinor\\_zero\\_of\\_pos](#)

[Read exact source](#)

LEVEL 3

R09

## mul\_chartFactor\_eq\_of\_orientedMaximalMinorsProportional

MathlibAnnex.Matrix.mul\_chartFactor\_eq\_of\_orientedMaximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[chartFactor](#),  
[rowCoordinates\\_eq\\_of\\_orientedMaximalMinorsProportional](#)

### USED BY IN THIS PROJECT

[existsUnique\\_factor\\_of\\_orientedMaximalMinorsProportional](#),  
[mul\\_chartFactor\\_eq\\_of\\_maximalMinorsProportional](#)

[Read exact source](#)

LEVEL 3

Project-wide support

## eventually\_mollify\_fderiv\_memLpOn\_compact

MathlibAnnex.Mollification.eventually\_mollify\_fderiv\_memLpOn\_compact

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mollify\\_contDiff](#)

### USED BY IN THIS PROJECT

[tendsto\\_eLpNorm\\_fderiv\\_mollify\\_sub](#)

[Read exact source](#)

LEVEL 3

R02

## maxSlice\_isCompact

MathlibAnnex.NonemptyCompacts.isCompact\_maxSlice

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maxSlice](#)

### USED BY IN THIS PROJECT

[refine](#)

[Read exact source](#)

**LEVEL 3**

R02

**maximizer\_mem\_maxSlice**

MathlibAnnex.NonemptyCompacts.maximizer\_mem\_maxSlice

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[maxSlice](#), [maximizer\\_mem](#)

**USED BY IN THIS PROJECT**

[maxSlice\\_nonempty](#)

[Read exact source](#)

**LEVEL 3**

R04

**componentFlux**

MathlibAnnex.Piola.componentFlux

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[cofactorRowField](#)

**USED BY IN THIS PROJECT**

[divergence\\_componentFlux\\_eq\\_det\\_sub](#),  
[support\\_componentFlux\\_subset](#)

[Read exact source](#)

**LEVEL 3**

R04

**sum\_hessian\_twoRowReplacement\_eq\_zero**

MathlibAnnex.Piola.sum\_hessian\_twoRowReplacement\_eq\_zero

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[det\\_twoRowReplacement\\_swap](#),  
[sum\\_symmetric\\_mul\\_antisymmetric\\_eq\\_zero](#)

**USED BY IN THIS PROJECT**

[sum\\_hessianCoordinate\\_twoRowReplacement\\_eq\\_zero](#)

[Read exact source](#)

**LEVEL 3**

Project-wide support

**referenceNorm\_le**

MathlibAnnex.PluckerRecovery.LinearCertificate.referenceNorm\_le

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[modelNorm\\_le\\_two](#)

**USED BY IN THIS PROJECT**

[exists\\_limitCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

**LEVEL 3**

R01

**radialExtension\_norm**

MathlibAnnex.Sphere.radialExtension\_norm

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[radialExtension\\_of\\_ne\\_zero](#), [radialExtension\\_zero](#)

**USED BY IN THIS PROJECT**

[radialExtension\\_lipschitz](#), [radialExtension\\_leftInverse](#)

[Read exact source](#)

**LEVEL 3**

R01

**radialExtension\_on\_sphere**

MathlibAnnex.Sphere.radialExtension\_on\_sphere

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[normalizeToSphere\\_unit](#), [radialExtension\\_of\\_ne\\_zero](#)

**USED BY IN THIS PROJECT**

[normalizedGenerator\\_mem\\_of\\_spherelsometry](#)

[Read exact source](#)

**LEVEL 3**

Project-wide support

**inner\_nonempty**`MathlibAnnex.WeakGradient.LocalBallData.inner_nonempty`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[center\\_mem\\_inner](#)**USED BY IN THIS PROJECT**[localBall\\_inner\\_measure\\_pos](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**middle\_subset\_open**`MathlibAnnex.WeakGradient.LocalBallData.middle_subset_open`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier\\_subset](#), [middle\\_subset\\_carrier](#)**USED BY IN THIS PROJECT**[translatedBumpField\\_carrier\\_subset](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**localMollification\_tendsto\_ae**`MathlibAnnex.WeakGradient.ae_tendsto_localMollification`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[localMollification](#), [shrinkingBump\\_ratio](#),  
[shrinkingBump\\_rOut\\_tendsto\\_zero](#)**USED BY IN THIS PROJECT**[exists\\_inner\\_convergencePoint](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**compactLocalization\_eq\_ae\_inner**`MathlibAnnex.WeakGradient.compactLocalization_eq_ae_inner`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[inner\\_subset\\_carrier](#), [inner\\_isOpen](#),  
[compactLocalization\\_of\\_mem](#)**USED BY IN THIS PROJECT**[exists\\_aeConstant\\_inner](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**divergence\_translatedBumpField**`MathlibAnnex.WeakGradient.divergence_translatedBumpField`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[translatedBumpField](#), [divergence\\_sub\\_smul](#)**USED BY IN THIS PROJECT**[localMollification\\_fderiv\\_apply\\_eq\\_zero](#)[Read exact source](#)**LEVEL 3**

Project-wide support

**integrableOn\_localBallCarrier**`MathlibAnnex.WeakGradient.integrableOn_localBallCarrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier\\_subset](#), [carrier\\_isCompact](#)**USED BY IN THIS PROJECT**[compactLocalization\\_integrable](#)[Read exact source](#)

**LEVEL 3**

R02

**allCoordinates\_separate**

MathlibAnnex.allCoordinates\_separate

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinate\\_mem\\_allCoordinates](#)**USED BY IN THIS PROJECT**[exists\\_common\\_pi](#)[Read exact source](#)**LEVEL 3**

R05

**constantRegion\_eq\_of\_preconnected**

MathlibAnnex.constantRegion\_eq\_of\_preconnected

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[constantRegion\\_isOpen](#), [constantRegion\\_compl\\_isOpen](#)**USED BY IN THIS PROJECT**[exists\\_aeConstantOn\\_of\\_local](#)[Read exact source](#)**LEVEL 3**

R02

**convexHull\_maxSlice\_eq\_supportFace**

MathlibAnnex.convexHull\_maxSlice\_eq\_supportFace

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[le\\_maximizer](#), [maxSlice](#), [supportFace](#)**USED BY IN THIS PROJECT**[refine\\_convexHull\\_eq](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 3**

R02

**maxValue\_eq\_of\_convexHull\_eq**

MathlibAnnex.maxValue\_eq\_of\_convexHull\_eq

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[maximizer\\_mem](#), [le\\_maxValue\\_of\\_mem\\_convexHull](#)**USED BY IN THIS PROJECT**[refine\\_convexHull\\_eq](#)[Read exact source](#)**Level 4** 34 declarations**LEVEL 4**

Project-wide support

**det\_eq\_domainSign\_mul\_abs**

MathlibAnnex.BilipschitzOrientation.det\_eq\_domainSign\_mul\_abs

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[det\\_fderiv\\_ne\\_zero](#), [domainJacobianSign](#)**USED BY IN THIS PROJECT**[signed\\_area\\_transfer](#)[Read exact source](#)**LEVEL 4**

Project-wide support

**targetJacobianSign\_locallyIntegrableOn**

MathlibAnnex.BilipschitzOrientation.locallyIntegrableOn\_targetJacobianSign

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_targetJacobianSign](#)**USED BY IN THIS PROJECT**[targetJacobianSign\\_ae\\_const](#)[Read exact source](#)

**LEVEL 4**

Project-wide support

**targetJacobianSign\_const\_is\_pm\_one**`MathlibAnnex.BilipschitzOrientation.targetJacobianSign_const_is_pm_one`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[AECOnstantOn](#), [abs\\_targetJacobianSign](#)**USED BY IN THIS PROJECT**[integral\\_det\\_fderiv\\_eq\\_signed\\_volume](#)[Read exact source](#)**LEVEL 4**

Project-wide support

**divergence\_eq\_zero\_of\_not\_mem\_carrier**`MathlibAnnex.CompactC1VectorField.divergence_eq_zero_of_not_mem_carrier`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[fderiv\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#), [divergence](#)**USED BY IN THIS PROJECT**[support\\_divergence\\_subset](#)[Read exact source](#)**LEVEL 4**

R03

**norm\_det\_sub\_le**`MathlibAnnex.ContinuousLinearMap.norm_det_sub_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_det\\_sub\\_le](#)**USED BY IN THIS PROJECT**[tendsto\\_integral\\_det\\_of\\_strongLn](#),  
[integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#)[Read exact source](#)**LEVEL 4**

Project-wide support

**selectedOutput**`MathlibAnnex.ContinuousLinearMap.selectedOutput`

Mathematical construction

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[norm\\_selectedOutputLinearMap\\_le](#)**USED BY IN THIS PROJECT**[norm\\_selectedOutput\\_le\\_one](#), [selectedOutput\\_apply](#),  
[selectedSquare](#)

▼ Show 2 more

[selectedSquareCLM](#), [selectedOutput\\_hasCompactSupport](#)[Read exact source](#)**LEVEL 4**

R07

**coordinateRow**`MathlibAnnex.DeterminantFrame.coordinateRow`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinateScale](#)**USED BY IN THIS PROJECT**[coordinateFrame](#), [coordinateRow\\_apply](#)[Read exact source](#)**LEVEL 4**

R07

**coordinateScale\_pos**`MathlibAnnex.DeterminantFrame.coordinateScale_pos`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[coordinateRowSize\\_nonneg](#), [coordinateScale](#)**USED BY IN THIS PROJECT**[coordinateFrame\\_mem](#)[Read exact source](#)

**LEVEL 4**

R07

**exists\_maximizingFrame**

MathlibAnnex.DeterminantFrame.exists\_maximizingFrame

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[frameDeterminant\\_continuous](#), [unitFrameSet\\_isCompact](#), [unitFrameSet\\_nonempty](#)

**USED BY IN THIS PROJECT**

[maximizingFrame](#)

[Open Declaration Card](#) [Read exact source](#)

**LEVEL 4**

R07

**replacementDeterminant\_eq\_cramerTranspose**

MathlibAnnex.DeterminantFrame.replacementDeterminant\_eq\_cramerTranspose

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[frameMatrix\\_replaceRow](#), [replacementDeterminant](#)

**USED BY IN THIS PROJECT**

[cramerReplacement](#)

[Read exact source](#)

**LEVEL 4**

Project-wide support

**closedUnitBall\_isCompact**

MathlibAnnex.EquivalentSeminorm.isCompact\_closedUnitBall

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[closedUnitBall\\_isBounded](#)

**USED BY IN THIS PROJECT**

[closedUnitBallVolume\\_pos](#), [derivativeGenerator\\_integrable\\_of\\_seminormLipschitz](#), [measure\\_image\\_unitBall\\_eq](#)

▼ Show 1 more

[radialExtension\\_integral\\_abs\\_det](#)

[Read exact source](#)

**LEVEL 4**

Project-wide support

**sphereIsometryEquiv**

MathlibAnnex.EquivalentSeminorm.sphereIsometryEquiv

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[dist\\_space\\_eq](#), [unitSphereEquiv](#)

**USED BY IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_coordinate\\_model](#)

[Read exact source](#)

**LEVEL 4**

Project-wide support

**configurationFinMap\_apply\_base**

MathlibAnnex.FiniteSup.Bridge.configurationFinMap\_apply\_base

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[configurationFinMap](#)

**USED BY IN THIS PROJECT**

[finMapConfiguration\\_configurationFinMap](#), [pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)

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**LEVEL 4**

Project-wide support

**configurationFinMap\_apply\_satellite**

MathlibAnnex.FiniteSup.Bridge.configurationFinMap\_apply\_satellite

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[configurationFinMap](#)

**USED BY IN THIS PROJECT**

[finMapConfiguration\\_configurationFinMap](#), [pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)

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LEVEL 4

Project-wide support

## configurationFinMap\_finMapConfiguration

MathlibAnnex.FiniteSup.Bridge.configurationFinMap\_finMapConfiguration

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationFinMap](#), [finMapConfiguration](#)

### USED BY IN THIS PROJECT

[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#)

[Read exact source](#)

LEVEL 4

Project-wide support

## configurationFinMap\_norm

MathlibAnnex.FiniteSup.Bridge.configurationFinMap\_norm

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationFinMap](#)

### USED BY IN THIS PROJECT

[configurationFinMap\\_norm\\_le\\_model](#)

[Read exact source](#)

LEVEL 4

R09

## MaximalMinorsProportional

MathlibAnnex.Matrix.MaximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#)

### USED BY IN THIS PROJECT

[orientedMaximalMinor\\_eq\\_of\\_orderEmbedding](#)

[Read exact source](#)

LEVEL 4

R09

## existsUnique\_factor\_of\_orientedMaximalMinorsProportional

MathlibAnnex.Matrix.existsUnique\_factor\_of\_orientedMaximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[chartFactor\\_unique](#), [det\\_chartFactor](#), [mul\\_chartFactor\\_eq\\_of\\_orientedMaximalMinorsProportional](#)

### USED BY IN THIS PROJECT

None in this Project

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LEVEL 4

R03

## maximalMinor\_mul

MathlibAnnex.Matrix.maximalMinor\_mul

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#), [maximalSubmatrix\\_mul](#)

### USED BY IN THIS PROJECT

[maximalMinors\\_mul](#), [derivativeGenerator\\_comp\\_linear\\_apply](#), [exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 4

R03

## maximalMinor\_ofOrderEmbedding

MathlibAnnex.Matrix.maximalMinor\_ofOrderEmbedding

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#), [maximalSubmatrix\\_ofOrderEmbedding](#)

### USED BY IN THIS PROJECT

[frameMap\\_injective\\_of\\_det\\_ne\\_zero](#), [orientedMaximalMinor\\_eq\\_of\\_orderEmbedding](#), [pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)

[Read exact source](#)

LEVEL 4

R03

## maximalMinor\_zero\_of\_pos

MathlibAnnex.Matrix.maximalMinor\_zero\_of\_pos

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#), [maximalSubmatrix\\_zero](#)

### USED BY IN THIS PROJECT

[maximalMinors\\_zero\\_of\\_pos](#)

[Read exact source](#)

LEVEL 4

R03

## maximalMinors

MathlibAnnex.Matrix.maximalMinors

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#)

### USED BY IN THIS PROJECT

[ballVolumeScaledMaximalMinors](#),  
[maximalMinors\\_continuous](#), [maximalMinors\\_mul](#)

### ▼ Show 3 more

[maximalMinors\\_smul](#), [maximalMinors\\_zero\\_of\\_pos](#),  
[pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)

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LEVEL 4

R03

## mulVec\_injective\_of\_maximalMinor\_ne\_zero

MathlibAnnex.Matrix.mulVec\_injective\_of\_maximalMinor\_ne\_zero

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor](#), [maximalSubmatrix\\_mulVec](#)

### USED BY IN THIS PROJECT

[frameMap\\_injective\\_of\\_det\\_ne\\_zero](#),  
[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 4

Project-wide support

## tendsto\_eLpNorm\_fderiv\_mollify\_sub

MathlibAnnex.Mollification.tendsto\_eLpNorm\_fderiv\_mollify\_sub

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[eventually\\_mollify\\_fderiv\\_memLpOn\\_compact](#)

### USED BY IN THIS PROJECT

[tendsto\\_integral\\_maximalMinor\\_mollify](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 4

R02

## maxSlice\_nonempty

MathlibAnnex.NonemptyCompacts.maxSlice\_nonempty

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizer\\_mem\\_maxSlice](#)

### USED BY IN THIS PROJECT

[refine](#)

[Read exact source](#)

LEVEL 4

R04

## sum\_hessianCoordinate\_twoRowReplacement\_eq\_zero

MathlibAnnex.Piola.sum\_hessianCoordinate\_twoRowReplacement\_eq\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[hessianCoordinate\\_comm](#),  
[sum\\_hessian\\_twoRowReplacement\\_eq\\_zero](#)

### USED BY IN THIS PROJECT

[piola\\_divergence\\_eq\\_zero\\_of\\_expansion](#)

[Read exact source](#)

LEVEL 4

R04

## support\_componentFlux\_subset

MathlibAnnex.Piola.support\_componentFlux\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[componentFlux](#)

### USED BY IN THIS PROJECT

[componentFlux\\_hasCompactSupport](#)

[Read exact source](#)

LEVEL 4

Project-wide support

## satellitePolynomial

MathlibAnnex.Satellite.satellitePolynomial

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[replacementDeterminant](#)

### USED BY IN THIS PROJECT

[abs\\_satellitePolynomial\\_le](#), [configurationPolynomial](#)

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LEVEL 4

R01

## radialExtension\_lipschitz

MathlibAnnex.Sphere.lipschitzWith\_radialExtension

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_norm](#), [scaled\\_normalize\\_dist\\_le\\_two](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#), [finrank\\_eq](#),  
[radialExtension\\_symm\\_lipschitz](#)

▼ Show 1 more

[radialExtension\\_integral\\_abs\\_det](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 4

R01

## radialExtension\_leftInverse

MathlibAnnex.Sphere.radialExtension\_leftInverse

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_norm](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_comp\\_linear\\_radial](#),  
[radialExtension\\_image\\_ball](#), [radialExtension\\_rightInverse](#)

[Open Declaration Card](#) [Read exact source](#)

**LEVEL 4**

Project-wide support

**compactLocalization\_integrable**`MathlibAnnex.WeakGradient.integrable_compactLocalization`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier\\_measurable](#), [compactLocalization](#),  
[integrableOn\\_localBallCarrier](#)**USED BY IN THIS PROJECT**[compactLocalization\\_locallyIntegrable](#)[Read exact source](#)**LEVEL 4**

Project-wide support

**localBall\_inner\_measure\_pos**`MathlibAnnex.WeakGradient.localBall_inner_measure_pos`

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[inner\\_nonempty](#), [inner\\_isOpen](#)**USED BY IN THIS PROJECT**[exists\\_inner\\_convergencePoint](#)[Read exact source](#)**LEVEL 4**

Project-wide support

**translatedBumpField\_carrier\_subset**`MathlibAnnex.WeakGradient.translatedBumpField_carrier_subset`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[carrier](#), [middle\\_subset\\_open](#),  
[translated\\_closedBall\\_subset\\_middle](#)**▼ Show 2 more**[shrinkingBump\\_rOut\\_le\\_half](#), [translatedBumpField](#)**USED BY IN THIS PROJECT**[weakIntegral\\_eq\\_localized\\_bumpIntegral](#)[Read exact source](#)**LEVEL 4**

R05

**exists\_aeConstantOn\_of\_local**`MathlibAnnex.exists_aeConstantOn_of_local`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[aeConstantOn\\_of\\_everywhere\\_local](#),  
[constantRegion\\_eq\\_of\\_preconnected](#)**USED BY IN THIS PROJECT**[exists\\_aeConstantOn\\_of\\_local\\_preconnected](#)[Open Declaration Card](#) [Read exact source](#)**Level 5** 33 declarations

LEVEL 5

Project-wide support

## signed\_area\_transfer

MathlibAnnex.BilipschitzOrientation.signed\_area\_transfer

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_domainJacobianSign](#), [det\\_eq\\_domainSign\\_mul\\_abs](#), [targetJacobianSign](#)

▼ Show 1 more

[target\\_exceptional\\_null](#)

### USED BY IN THIS PROJECT

[targetJacobianSign\\_weakDivergenceZero](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 5

Project-wide support

## support\_divergence\_subset

MathlibAnnex.CompactC1VectorField.support\_divergence\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[divergence\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 5

Project-wide support

## norm\_selectedOutput\_le\_one

MathlibAnnex.ContinuousLinearMap.norm\_selectedOutput\_le\_one

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput](#)

### USED BY IN THIS PROJECT

[norm\\_selectedSquare\\_le](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## selectedOutput\_apply

MathlibAnnex.ContinuousLinearMap.selectedOutput\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 5

Project-wide support

## selectedSquare

MathlibAnnex.ContinuousLinearMap.selectedSquare

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput](#)

### USED BY IN THIS PROJECT

[det\\_selectedSquare](#), [norm\\_selectedSquare\\_le](#), [selectedSquareCLM\\_apply](#)

▼ Show 2 more

[selectedSquare\\_sub](#), [maximalMinorIntegrand](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## selectedSquareCLM

MathlibAnnex.ContinuousLinearMap.selectedSquareCLM

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput](#)

### USED BY IN THIS PROJECT

[selectedSquareCLM\\_apply](#), [selectedSquare\\_sub](#), [integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#)

[Read exact source](#)

LEVEL 5

R07

## coordinateFrame

MathlibAnnex.DeterminantFrame.coordinateFrame

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Frame](#), [coordinateRow](#)

### USED BY IN THIS PROJECT

[coordinateFrame\\_mem](#), [frameMatrix\\_coordinateFrame](#)

[Read exact source](#)

LEVEL 5

R07

## coordinateRow\_apply

MathlibAnnex.DeterminantFrame.coordinateRow\_apply

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateRow](#)

### USED BY IN THIS PROJECT

[abs\\_inverse\\_mulVec\\_apply\\_le](#),  
[frameMatrix\\_coordinateFrame](#)

[Read exact source](#)

LEVEL 5

R07

## frameMap\_injective\_of\_det\_ne\_zero

MathlibAnnex.DeterminantFrame.frameMap\_injective\_of\_det\_ne\_zero

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameCoordinates\\_eq\\_mulVec](#), [frameDeterminant](#),  
[frameMap](#)

#### ▼ Show 2 more

[maximalMinor\\_ofOrderEmbedding](#),  
[mulVec\\_injective\\_of\\_maximalMinor\\_ne\\_zero](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 5

R07

## maximizingFrame

MathlibAnnex.DeterminantFrame.maximizingFrame

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists\\_maximizingFrame](#)

### USED BY IN THIS PROJECT

[abs\\_frameDeterminant\\_le\\_maximizingFrame](#),  
[determinantMaximum](#), [maximizingFrame\\_mem](#)

[Read exact source](#)

LEVEL 5

R07

## cramerReplacement

MathlibAnnex.DeterminantFrame.sum\_mul\_replacementDeterminant

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[functional\\_apply\\_eq\\_sum](#),  
[replacementDeterminant\\_eq\\_cramerTranspose](#)

### USED BY IN THIS PROJECT

[abs\\_inverse\\_mulVec\\_apply\\_le](#),  
[absoluteMaximizer\\_satellite\\_norms\\_preimage](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## closedUnitBallVolume\_pos

MathlibAnnex.EquivalentSeminorm.closedUnitBallVolume\_pos

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBallVolume](#), [closedUnitBall\\_isCompact](#),  
[zero\\_mem\\_interior\\_closedUnitBall](#)

### USED BY IN THIS PROJECT

[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#), [derivativeAverage\\_comp\\_linear\\_radial\\_apply](#),  
[derivativeAverage\\_mem\\_body](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## configurationFinMap\_norm\_le\_model

MathlibAnnex.FiniteSup.Bridge.configurationFinMap\_norm\_le\_model

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationFinMap\\_norm](#),  
[configurationMap\\_norm\\_le\\_model](#)

### USED BY IN THIS PROJECT

[configuration\\_contraction\\_iff](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## finMapConfiguration\_configurationFinMap

MathlibAnnex.FiniteSup.Bridge.finMapConfiguration\_configurationFinMap

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[configurationFinMap\\_apply\\_base](#),  
[configurationFinMap\\_apply\\_satellite](#), [finMapConfiguration](#)

### USED BY IN THIS PROJECT

[configuration\\_contraction\\_iff](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## ballVolumeScaledMaximalMinors

MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBallVolume](#), [maximalMinors](#)

### USED BY IN THIS PROJECT

[PluckerGeneratorGood](#),  
[ballVolumeScaledMaximalMinors\\_zero\\_of\\_pos](#),  
[ballVolumeScaledMaximalMinors\\_continuous](#)

### ▼ Show 3 more

[pairing\\_ballVolumeScaledMaximalMinors](#),  
[derivativeGenerator](#), [generators](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 5

R03

## maximalMinors\_continuous

MathlibAnnex.Matrix.continuous\_maximalMinors

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinors](#)

### USED BY IN THIS PROJECT

[ballVolumeScaledMaximalMinors\\_continuous](#)

[Read exact source](#)

LEVEL 5

R03

## maximalMinors\_mul

MathlibAnnex.Matrix.maximalMinors\_mul

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor\\_mul](#), [maximalMinors](#)

### USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

LEVEL 5

R03

## maximalMinors\_smul

MathlibAnnex.Matrix.maximalMinors\_smul

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinors](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 5

R03

## maximalMinors\_zero\_of\_pos

MathlibAnnex.Matrix.maximalMinors\_zero\_of\_pos

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor\\_zero\\_of\\_pos](#), [maximalMinors](#)

### USED BY IN THIS PROJECT

[ballVolumeScaledMaximalMinors\\_zero\\_of\\_pos](#)

[Read exact source](#)

LEVEL 5

R09

## orientedMaximalMinor\_eq\_of\_orderEmbedding

MathlibAnnex.Matrix.orientedMaximalMinor\_eq\_of\_orderEmbedding

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[MaximalMinorsProportional](#),  
[maximalMinor\\_ofOrderEmbedding](#), [orientedMaximalMinor](#)

### USED BY IN THIS PROJECT

[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#)

[Read exact source](#)

LEVEL 5

R04

## tendsto\_integral\_det\_of\_strongLn

MathlibAnnex.MeasureTheory.tendsto\_integral\_det\_of\_strongLn

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[norm\\_det\\_sub\\_le](#), [StrongLnOperatorField](#)

### USED BY IN THIS PROJECT

[tendsto\\_integral\\_maximalMinor\\_mollify](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 5

R02

## refine

MathlibAnnex.NonemptyCompacts.refine

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maxSlice\\_isCompact](#), [maxSlice\\_nonempty](#)

### USED BY IN THIS PROJECT

[mem\\_refine](#), [lexicographicRefine](#), [refine\\_convexHull\\_eq](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## selectedOutput\_hasCompactSupport

MathlibAnnex.NullLagrangian.hasCompactSupport\_selectedOutput

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDif f](#)

[Read exact source](#)

LEVEL 5

R04

## componentFlux\_hasCompactSupport

MathlibAnnex.Piola.hasCompactSupport\_componentFlux

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[support\\_componentFlux\\_subset](#)

### USED BY IN THIS PROJECT

[integral\\_det\\_singleOutputPerturb\\_sub\\_eq\\_zero](#)

[Read exact source](#)

LEVEL 5

R04

## piola\_divergence\_eq\_zero\_of\_expansion

MathlibAnnex.Piola.piola\_divergence\_eq\_zero\_of\_expansion

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[cofactorRowField](#),  
[sum\\_hessianCoordinate\\_twoRowReplacement\\_eq\\_zero](#)

### USED BY IN THIS PROJECT

[divergence\\_cofactorRowField\\_eq\\_zero](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## measure\_image\_unitBall\_eq

MathlibAnnex.PluckerRecovery.LimitCertificate.measure\_image\_unitBall\_eq

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBallVolume](#), [closedUnitBall\\_isCompact](#),  
[LimitCertificate](#)

### USED BY IN THIS PROJECT

[image\\_unitBall\\_eq](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## configurationPolynomial

MathlibAnnex.Satellite.configurationPolynomial

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[SatelliteConfiguration](#), [satellitePolynomial](#)

### USED BY IN THIS PROJECT

[orientedSatelliteSupport](#),  
[pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#),  
[absoluteMaximizer\\_base\\_nearMax](#)

▼ Show 2 more

[absoluteMaximizer\\_satellite\\_norms\\_preimage](#),  
[maxSatelliteConfiguration](#)

[Read exact source](#)

LEVEL 5

R01

## radialExtension\_symm\_lipschitz

MathlibAnnex.Sphere.lipschitzWith\_radialExtension\_symm

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_lipschitz](#)

### USED BY IN THIS PROJECT

[radialExtensionHomeomorph](#)

[Read exact source](#)

LEVEL 5

Project-wide support

## radialExtension\_image\_ball

MathlibAnnex.Sphere.radialExtension\_image\_ball

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[Space\\_eq\\_zero\\_of\\_apply\\_eq\\_zero](#),  
[radialExtension\\_leftInverse](#)

### USED BY IN THIS PROJECT

[radialExtension\\_integral\\_abs\\_det](#)

[Read exact source](#)

LEVEL 5

R01

## radialExtension\_rightInverse

MathlibAnnex.Sphere.radialExtension\_rightInverse

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_leftInverse](#)

### USED BY IN THIS PROJECT

[radialExtensionEquiv](#), [radialExtension\\_surjective](#)

[Read exact source](#)

**LEVEL 5**

Project-wide support

**compactLocalization\_locallyIntegrable**`MathlibAnnex.WeakGradient.locallyIntegrable_compactLocalization`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[compactLocalization\\_integrable](#)**USED BY IN THIS PROJECT**[localMollification\\_contDiff](#), [exists\\_inner\\_convergencePoint](#), [localMollification\\_fderiv\\_apply\\_eq\\_zero](#)[Read exact source](#)**LEVEL 5**

Project-wide support

**weakIntegral\_eq\_localized\_bumpIntegral**`MathlibAnnex.WeakGradient.weakIntegral_eq_localized_bumpIntegral`

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[fderiv\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#), [WeakDivergenceZero](#), [compactLocalization](#)**▼ Show 1 more**[translatedBumpField\\_carrier\\_subset](#)**USED BY IN THIS PROJECT**[localMollification\\_fderiv\\_apply\\_eq\\_zero](#)[Read exact source](#)**LEVEL 5**

R05

**exists\_aeConstantOn\_of\_local\_preconnected**`MathlibAnnex.exists_aeConstantOn_of_local_preconnected`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[exists\\_aeConstantOn\\_of\\_local](#)**USED BY IN THIS PROJECT**[exists\\_aeConstantOn](#)[Read exact source](#)**Level 6** 31 declarations**LEVEL 6**

Project-wide support

**det\_selectedSquare**`MathlibAnnex.ContinuousLinearMap.det_selectedSquare`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[selectedSquare](#), [maximalMinor](#)**USED BY IN THIS PROJECT**[topMinor\\_difference\\_eq\\_zero\\_of\\_not\\_mem\\_tsupport](#)[Read exact source](#)**LEVEL 6**

Project-wide support

**norm\_selectedSquare\_le**`MathlibAnnex.ContinuousLinearMap.norm_selectedSquare_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[norm\\_selectedOutput\\_le\\_one](#), [selectedSquare](#)**USED BY IN THIS PROJECT**[tendsto\\_integral\\_maximalMinor\\_mollify](#)[Read exact source](#)

LEVEL 6

Project-wide support

## selectedSquareCLM\_apply

MathlibAnnex.ContinuousLinearMap.selectedSquareCLM\_apply

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedSquare](#), [selectedSquareCLM](#)

### USED BY IN THIS PROJECT

[maximalMinorIntegrand\\_mollify\\_continuous](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## selectedSquare\_sub

MathlibAnnex.ContinuousLinearMap.selectedSquare\_sub

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedSquare](#), [selectedSquareCLM](#)

### USED BY IN THIS PROJECT

[tendsto\\_integral\\_maximalMinor\\_mollify](#)

[Read exact source](#)

LEVEL 6

R07

## abs\_frameDeterminant\_le\_maximizingFrame

MathlibAnnex.DeterminantFrame.abs\_frameDeterminant\_le\_maximizingFrame

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizingFrame](#)

### USED BY IN THIS PROJECT

[abs\\_replacementDeterminant\\_le\\_maximum](#),  
[determinantMaximum\\_pos](#)

[Read exact source](#)

LEVEL 6

R07

## coordinateFrame\_mem

MathlibAnnex.DeterminantFrame.coordinateFrame\_mem

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateFrame](#), [coordinateScale\\_pos](#), [mem\\_unitFrameSet](#)

### ▼ Show 1 more

[norm\\_rawCoordinateRow\\_le\\_size](#)

### USED BY IN THIS PROJECT

[abs\\_inverse\\_mulVec\\_apply\\_le](#), [determinantMaximum\\_pos](#)

[Read exact source](#)

LEVEL 6

R07

## determinantMaximum

MathlibAnnex.DeterminantFrame.determinantMaximum

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximizingFrame](#)

### USED BY IN THIS PROJECT

[abs\\_replacementDeterminant\\_le\\_maximum](#),  
[coordinateInverseFactor](#), [determinantMaximum\\_pos](#)

### ▼ Show 3 more

[nearMaxFrames](#), [CoefficientDetectsUnit](#), [satelliteBudget](#)

[Read exact source](#)

LEVEL 6

R07

## frameMatrix\_coordinateFrame

MathlibAnnex.DeterminantFrame.frameMatrix\_coordinateFrame

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateFrame](#), [coordinateRow\\_apply](#), [frameMatrix](#)

### USED BY IN THIS PROJECT

[frameDeterminant\\_coordinateFrame](#)

[Read exact source](#)

**LEVEL 6**

R07

**maximizingFrame\_mem**`MathlibAnnex.DeterminantFrame.maximizingFrame_mem`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[maximizingFrame](#)**USED BY IN THIS PROJECT**[nearMaxFrames\\_nonempty](#),  
[absoluteMaximizer\\_base\\_nearMax](#)[Read exact source](#)**LEVEL 6**

Project-wide support

**PluckerGeneratorGood**`MathlibAnnex.FiniteRecovery.PluckerGeneratorGood`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[IsContraction](#), [FinMapAlmostIsometric](#),  
[ballVolumeScaledMaximalMinors](#)**USED BY IN THIS PROJECT**[targetRawSupportMaximizer\\_good](#)[Read exact source](#)**LEVEL 6**

Project-wide support

**configuration\_contraction\_iff**`MathlibAnnex.FiniteSup.Bridge.configuration_contraction_iff`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[configurationFinMap\\_norm\\_le\\_model](#),  
[finMapConfiguration\\_configurationFinMap](#),  
[finMapConfiguration\\_mem](#)**USED BY IN THIS PROJECT**[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#)[Read exact source](#)**LEVEL 6**

Project-wide support

**ballVolumeScaledMaximalMinors\_zero\_of\_pos**`MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors_zero_of_pos`

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ballVolumeScaledMaximalMinors](#),  
[maximalMinors\\_zero\\_of\\_pos](#)**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

LEVEL 6

Project-wide support

## ballVolumeScaledMaximalMinors\_continuous

MathlibAnnex.Matrix.continuous\_ballVolumeScaledMaximalMinors

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ballVolumeScaledMaximalMinors](#),  
[maximalMinors\\_continuous](#)

### USED BY IN THIS PROJECT

[derivativeGenerator\\_integrableOn\\_compact](#),  
[generators\\_isCompact](#)

[Read exact source](#)

LEVEL 6

R09

## orientedMaximalMinorsProportional\_of\_maximalMinorsProportional

MathlibAnnex.Matrix.orientedMaximalMinorsProportional\_of\_maximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[OrientedMaximalMinorsProportional](#),  
[exists\\_orderEmbedding\\_perm\\_of\\_injective](#),  
[orientedMaximalMinor\\_eq\\_of\\_orderEmbedding](#)

### ▼ Show 2 more

[orientedMaximalMinor\\_eq\\_zero\\_of\\_not\\_injective](#),  
[orientedMaximalMinor\\_permute](#)

### USED BY IN THIS PROJECT

[mul\\_chartFactor\\_eq\\_of\\_maximalMinorsProportional](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## pairing\_ballVolumeScaledMaximalMinors

MathlibAnnex.Matrix.pairing\_ballVolumeScaledMaximalMinors

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ballVolumeScaledMaximalMinors](#)

### USED BY IN THIS PROJECT

[pluckerFunctional\\_normalized\\_configurationFinMap](#)

[Read exact source](#)

LEVEL 6

R02

## mem\_refine

MathlibAnnex.NonemptyCompacts.mem\_refine

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[refine](#)

### USED BY IN THIS PROJECT

[lexicographicRefine\\_subset](#)

[Read exact source](#)

**LEVEL 6**

Project-wide support

**maximalMinorIntegrand**`MathlibAnnex.NullLagrangian.maximalMinorIntegrand`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[selectedSquare](#)**USED BY IN THIS PROJECT**[maximalMinorIntegrand\\_mollify\\_continuous](#),  
[integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#),  
[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDif](#)  
[f](#)**▼ Show 2 more**[tendsto\\_integral\\_maximalMinor\\_mollify](#),  
[topMinor\\_difference\\_eq\\_zero\\_of\\_not\\_mem\\_tsupport](#)[Read exact source](#)**LEVEL 6**

R04

**divergence\_cofactorRowField\_eq\_zero**`MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[basisRow\\_apply](#), [piola\\_divergence\\_eq\\_zero\\_of\\_expansion](#)**USED BY IN THIS PROJECT**[divergence\\_componentFlux\\_eq\\_det\\_sub](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 6**

Project-wide support

**derivativeGenerator**`MathlibAnnex.Plucker.derivativeGenerator`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ballVolumeScaledMaximalMinors](#)**USED BY IN THIS PROJECT**[derivativeAverage](#), [derivativeGenerator\\_comp\\_linear\\_apply](#),  
[derivativeGenerator\\_mem\\_raw](#)**▼ Show 1 more**[derivativeGenerator\\_integrableOn\\_compact](#)[Read exact source](#)**LEVEL 6**

Project-wide support

**generators**`MathlibAnnex.PluckerBody.generators`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[IsContraction](#), [ballVolumeScaledMaximalMinors](#)**USED BY IN THIS PROJECT**[derivativeGenerator\\_mem\\_raw](#), [body](#),  
[generators\\_eq\\_image\\_union](#)**▼ Show 2 more**[generators\\_neg](#), [generators\\_nonempty](#)[Read exact source](#)

LEVEL 6

Project-wide support

## image\_unitBall\_eq

MathlibAnnex.PluckerRecovery.LimitCertificate.image\_unitBall\_eq

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[measure\\_image\\_unitBall\\_eq](#), [measure\\_lt](#)

### USED BY IN THIS PROJECT

[exists\\_linearIsometryEquiv\\_of\\_pluckerBodies\\_eq](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 6

Project-wide support

## orientedSatelliteSupport

MathlibAnnex.PluckerSupport.orientedSatelliteSupport

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[satellitePluckerCoefficients](#), [configurationPolynomial](#)

### USED BY IN THIS PROJECT

[orientedSatelliteSupport\\_configuration](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## pluckerPairing\_satelliteCoefficients\_maximalMinors

MathlibAnnex.PluckerSupport.pluckerPairing\_satelliteCoefficients\_maximalMinors

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameMatrix\\_replaceRow](#), [configurationFinMap\\_apply\\_base](#), [configurationFinMap\\_apply\\_satellite](#)

### ▼ Show 4 more

[maximalMinor\\_ofOrderEmbedding](#), [maximalMinors](#), [satellitePluckerCoefficients](#), [configurationPolynomial](#)

### USED BY IN THIS PROJECT

[pluckerFunctional\\_normalized\\_configurationFinMap](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 6

Project-wide support

## maxSatelliteConfiguration

MathlibAnnex.Satellite.maxSatelliteConfiguration

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameDeterminant\\_continuous](#), [configurationPolynomial](#), [satelliteConfigurationSet](#)

### USED BY IN THIS PROJECT

[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#), [maxNetConfigurationMap\\_lower\\_of\\_ne\\_zero](#), [exists\\_goodNetSatelliteMaximizer](#)

[Read exact source](#)

LEVEL 6

R01

## radialExtensionEquiv

MathlibAnnex.Sphere.radialExtensionEquiv

Rigidity or equivalence result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_rightInverse](#)

### USED BY IN THIS PROJECT

[radialExtensionHomeomorph](#)

[Read exact source](#)

LEVEL 6

R01

## radialExtension\_surjective

MathlibAnnex.Sphere.radialExtension\_surjective

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_rightInverse](#)

### USED BY IN THIS PROJECT

[finrank\\_eq](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## localMollification\_contDiff

MathlibAnnex.WeakGradient.contDiff\_localMollification

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[localMollification](#), [compactLocalization\\_locallyIntegrable](#)

### USED BY IN THIS PROJECT

[exists\\_localMollification\\_constant](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## exists\_inner\_convergencePoint

MathlibAnnex.WeakGradient.exists\_inner\_convergencePoint

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[localMollification\\_tendsto\\_ae](#), [localBall\\_inner\\_measure\\_pos](#), [compactLocalization\\_locallyIntegrable](#)

### USED BY IN THIS PROJECT

[compactLocalization\\_aeConstant\\_inner](#)

[Read exact source](#)

LEVEL 6

Project-wide support

## localMollification\_fderiv\_apply\_eq\_zero

MathlibAnnex.WeakGradient.localMollification\_fderiv\_apply\_eq\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[divergence\\_translatedBumpField](#), [localMollification\\_fderiv\\_apply](#), [compactLocalization\\_locallyIntegrable](#)

### ▼ Show 1 more

[weakIntegral\\_eq\\_localized\\_bumpIntegral](#)

### USED BY IN THIS PROJECT

[localMollification\\_fderiv\\_eq\\_zero](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 6

R02

## lexicographicRefine

MathlibAnnex.lexicographicRefine

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[refine](#)

### USED BY IN THIS PROJECT

[lexicographicRefine\\_convexHull\\_eq](#), [lexicographicRefine\\_subset](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 6

R02

## refine\_convexHull\_eq

MathlibAnnex.refine\_convexHull\_eq

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[refine\\_convexHull\\_maxSlice\\_eq\\_supportFace](#), [maxValue\\_eq\\_of\\_convexHull\\_eq](#)

### USED BY IN THIS PROJECT

[lexicographicRefine\\_convexHull\\_eq](#)

[Read exact source](#)

LEVEL 7

R07

abs\_replacementDeterminant\_le\_maximum

MathlibAnnex.DeterminantFrame.abs\_replacementDeterminant\_le\_maximum

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_frameDeterminant\\_le\\_maximizingFrame](#),  
[determinantMaximum](#), [replaceRow\\_mem\\_unitFrameSet](#)

▼ Show 1 more

[replacementDeterminant](#)

USED BY IN THIS PROJECT

[abs\\_cramerNumerator\\_le](#), [abs\\_satelliteTerms\\_le\\_budget](#)

[Read exact source](#)

LEVEL 7

R07

coordinateInverseFactor

MathlibAnnex.DeterminantFrame.coordinateInverseFactor

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateScale](#), [determinantMaximum](#)

USED BY IN THIS PROJECT

[inverseBoundConstant](#)

[Read exact source](#)

LEVEL 7

R07

frameDeterminant\_coordinateFrame

MathlibAnnex.DeterminantFrame.frameDeterminant\_coordinateFrame

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameDeterminant](#), [frameMatrix\\_coordinateFrame](#)

USED BY IN THIS PROJECT

[determinantMaximum\\_pos](#)

[Read exact source](#)

LEVEL 7

R07

nearMaxFrames

MathlibAnnex.DeterminantFrame.nearMaxFrames

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT

[determinantMaximum](#)

USED BY IN THIS PROJECT

[bound\\_sub](#), [nearMaxFrames\\_isCompact](#),  
[nearMaxFrame\\_abs\\_det\\_lower](#)

▼ Show 1 more

[nearMaxFrames\\_nonempty](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 7

R09

## mul\_chartFactor\_eq\_of\_maximalMinorsProportional

MathlibAnnex.Matrix.mul\_chartFactor\_eq\_of\_maximalMinorsProportional

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mul\\_chartFactor\\_eq\\_of\\_orientedMaximalMinorsProportional](#),  
[orientedMaximalMinorsProportional\\_of\\_maximalMinorsProportional](#)

### USED BY IN THIS PROJECT

[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## maximalMinorIntegrand\_mollify\_continuous

MathlibAnnex.NullLagrangian.continuous\_maximalMinorIntegrand\_mollify

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedSquareCLM\\_apply\\_mollify\\_contDiff](#),  
[maximalMinorIntegrand](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## integrableOn\_maximalMinor\_fderiv\_of\_lipschitzWith

MathlibAnnex.NullLagrangian.integrableOn\_maximalMinor\_fderiv\_of\_lipschitzWith

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[norm\\_det\\_sub\\_le\\_selectedSquareCLM](#),  
[lipschitz\\_fderiv\\_memLpOn\\_compact](#)

#### ▼ Show 1 more

[maximalMinorIntegrand](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#),  
[radialExtension\\_integral\\_abs\\_det](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## tendsto\_integral\_maximalMinor\_mollify

MathlibAnnex.NullLagrangian.tendsto\_integral\_maximalMinor\_mollify

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[norm\\_selectedSquare\\_le\\_selectedSquare\\_sub](#),  
[tendsto\\_integral\\_det\\_of\\_strongLn](#)

#### ▼ Show 3 more

[lipschitz\\_fderiv\\_memLpOn\\_compact](#),  
[tendsto\\_eLpNorm\\_fderiv\\_mollify\\_sub](#),  
[maximalMinorIntegrand](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## topMinor\_difference\_eq\_zero\_of\_not\_m\_tsupport

MathlibAnnex.NullLagrangian.topMinor\_difference\_eq\_zero\_of\_not\_mem\_tsupport

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[det\\_selectedSquare](#), [maximalMinorIntegrand](#)

### USED BY IN THIS PROJECT

[integral\\_topMinor\\_difference\\_eq\\_setIntegral](#)

[Read exact source](#)

LEVEL 7

R04

## divergence\_componentFlux\_eq\_det\_sub

MathlibAnnex.Piola.divergence\_componentFlux\_eq\_det\_sub

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[componentFlux](#),  
[det\\_updateRow\\_eq\\_sum\\_mul\\_cofactorRow](#),  
[divergence\\_cofactorRowField\\_eq\\_zero](#)

### ▼ Show 1 more

[singleOutputPerturb](#)

### USED BY IN THIS PROJECT

[integral\\_det\\_singleOutputPerturb\\_sub\\_eq\\_zero](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 7

Project-wide support

## derivativeAverage

MathlibAnnex.Plucker.derivativeAverage

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[derivativeGenerator](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#),  
[derivativeAverage\\_mem\\_body](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 7

Project-wide support

## derivativeGenerator\_comp\_linear\_apply

MathlibAnnex.Plucker.derivativeGenerator\_comp\_linear\_apply

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[maximalMinor\\_mul](#), [derivativeGenerator](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## derivativeGenerator\_mem\_raw

MathlibAnnex.Plucker.derivativeGenerator\_mem\_raw

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[derivativeGenerator](#), [generators](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_mem\\_body](#),  
[derivativeGenerator\\_integrable\\_of\\_seminormLipschitz](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## derivativeGenerator\_integrableOn\_compact

MathlibAnnex.Plucker.integrableOn\_derivativeGenerator\_compact

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[ballVolumeScaledMaximalMinors\\_continuous](#),  
[derivativeGenerator](#)

### USED BY IN THIS PROJECT

[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## body

MathlibAnnex.PluckerBody.body

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[generators](#)

USED BY IN THIS PROJECT  
[exists\\_almostIsometryMatch\\_at\\_satelliteDimension](#),  
[derivativeAverage\\_mem\\_body](#), [body\\_neg](#)

▼ Show 1 more

[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 7

Project-wide support

## generators\_eq\_image\_union

MathlibAnnex.PluckerBody.generators\_eq\_image\_union

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[contractionSet](#), [generators](#)

USED BY IN THIS PROJECT  
[generators\\_isCompact](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## generators\_neg

MathlibAnnex.PluckerBody.generators\_neg

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[generators](#)

USED BY IN THIS PROJECT  
[body\\_neg](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## generators\_nonempty

MathlibAnnex.PluckerBody.generators\_nonempty

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[zero\\_isContraction](#), [generators](#)

USED BY IN THIS PROJECT  
[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## pluckerFunctional\_normalized\_configurationFinMap

MathlibAnnex.PluckerSupport.pluckerFunctional\_normalized\_configurationFinMap

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[pairing\\_ballVolumeScaledMaximalMinors](#),  
[pluckerPairing\\_satelliteCoefficients\\_maximalMinors](#)

USED BY IN THIS PROJECT  
[orientedSatelliteSupport\\_configuration](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## CoefficientDetectsUnit

MathlibAnnex.Satellite.CoefficientDetectsUnit

Mathematical construction

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[NearMaxInverseBound](#), [determinantMaximum](#), [Space](#)

▼ Show 1 more

[eq\\_zero\\_of\\_apply\\_eq\\_zero](#)

USED BY IN THIS PROJECT  
[absoluteMaximizer\\_satellites\\_almost\\_norm](#),  
[finiteCoefficientNet\\_detectsUnit](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## satelliteBudget

MathlibAnnex.Satellite.satelliteBudget

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[determinantMaximum](#), [Space](#), [eq\\_zero\\_of\\_apply\\_eq\\_zero](#)

### USED BY IN THIS PROJECT

[abs\\_satelliteTerms\\_le\\_budget](#),  
[exists\\_weight\\_dominating\\_budget](#)

[Read exact source](#)

LEVEL 7

R01

## finrank\_eq

MathlibAnnex.Sphere.finrank\_eq

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_lipschitz](#), [radialExtension\\_surjective](#)

### USED BY IN THIS PROJECT

[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 7

R01

## radialExtensionHomeomorph

MathlibAnnex.Sphere.radialExtensionHomeomorph

Mathematical construction

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[radialExtension\\_symm\\_lipschitz](#), [radialExtensionEquiv](#)

### USED BY IN THIS PROJECT

[finiteDimensional\\_codomain](#)

[Read exact source](#)

LEVEL 7

Project-wide support

## localMollification\_fderiv\_eq\_zero

MathlibAnnex.WeakGradient.localMollification\_fderiv\_eq\_zero

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[localMollification\\_fderiv\\_apply\\_eq\\_zero](#)

### USED BY IN THIS PROJECT

[exists\\_localMollification\\_constant](#)

[Read exact source](#)

LEVEL 7

R02

## lexicographicRefine\_convexHull\_eq

MathlibAnnex.lexicographicRefine\_convexHull\_eq

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[lexicographicRefine](#), [refine\\_convexHull\\_eq](#)

### USED BY IN THIS PROJECT

[exists\\_common\\_of\\_convexHull\\_eq](#)

[Read exact source](#)

LEVEL 7

R02

## lexicographicRefine\_subset

MathlibAnnex.lexicographicRefine\_subset

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[mem\\_refine](#), [lexicographicRefine](#)

### USED BY IN THIS PROJECT

[lexicographicRefine\\_agreesOn](#)

[Read exact source](#)

## Level 8 18 declarations

LEVEL 8

R07

## bound\_sub

MathlibAnnex.DeterminantFrame.NearMaxInverseBound.bound\_sub

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[NearMaxInverseBound](#), [coordinateEquiv](#), [nearMaxFrames](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 8

R07

## abs\_cramerNumerator\_le

MathlibAnnex.DeterminantFrame.abs\_cramerNumerator\_le

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_replacementDeterminant\\_le\\_maximum](#)

### USED BY IN THIS PROJECT

[abs\\_inverse\\_mulVec\\_apply\\_le](#)

[Read exact source](#)

LEVEL 8

R07

## determinantMaximum\_pos

MathlibAnnex.DeterminantFrame.determinantMaximum\_pos

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_frameDeterminant\\_le\\_maximizingFrame](#),  
[coordinateFrame\\_mem](#), [determinantMaximum](#)

### ▼ Show 1 more

[frameDeterminant\\_coordinateFrame](#)

### USED BY IN THIS PROJECT

[inverseBoundConstant\\_bound](#), [inverseBoundConstant\\_pos](#),  
[exists\\_weight\\_dominating\\_budget](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 8

R07

## inverseBoundConstant

MathlibAnnex.DeterminantFrame.inverseBoundConstant

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[coordinateInverseFactor](#), [inverseCoordinateMap](#)

### USED BY IN THIS PROJECT

[inverseBoundConstant\\_bound](#), [inverseBoundConstant\\_pos](#)

[Read exact source](#)

LEVEL 8

R07

## nearMaxFrames\_isCompact

MathlibAnnex.DeterminantFrame.isCompact\_nearMaxFrames

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[nearMaxFrames](#)

### USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 8

R07

## nearMaxFrame\_abs\_det\_lower

MathlibAnnex.DeterminantFrame.nearMaxFrame\_abs\_det\_lower

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[nearMaxFrames](#)

### USED BY IN THIS PROJECT

[nearMaxFrame\\_det\\_ne\\_zero](#)

[Read exact source](#)

**LEVEL 8**

R07

**nearMaxFrames\_nonempty**

MathlibAnnex.DeterminantFrame.nearMaxFrames\_nonempty

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[maximizingFrame\\_mem](#), [nearMaxFrames](#)

**USED BY IN THIS PROJECT**

None in this Project

[Read exact source](#)

**LEVEL 8**

Project-wide support

**integral\_topMinor\_difference\_eq\_setIntegral**

MathlibAnnex.NullLagrangian.integral\_topMinor\_difference\_eq\_setIntegral

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[topMinor\\_difference\\_eq\\_zero\\_of\\_not\\_mem\\_tsupport](#)

**USED BY IN THIS PROJECT**

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

**LEVEL 8**

R04

**integral\_det\_singleOutputPerturb\_sub\_eq\_zero**

MathlibAnnex.Piola.integral\_det\_singleOutputPerturb\_sub\_eq\_zero

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[divergence\\_componentFlux\\_eq\\_det\\_sub](#),  
[componentFlux\\_hasCompactSupport](#),  
[integral\\_coordinateDivergence\\_eq\\_zero\\_of\\_contDiff\\_hasCompactSupport](#)

**USED BY IN THIS PROJECT**

[integral\\_det\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)

[Read exact source](#)

**LEVEL 8**

Project-wide support

**derivativeAverage\_comp\_linear\_radial\_apply**

MathlibAnnex.Plucker.derivativeAverage\_comp\_linear\_radial\_apply

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[closedUnitBallVolume\\_pos](#), [dist\\_space\\_eq](#),  
[derivativeAverage](#)

**▼ Show 3 more**

[derivativeGenerator\\_comp\\_linear\\_apply](#),  
[derivativeGenerator\\_integrableOn\\_compact](#),  
[radialExtension\\_lipschitz](#)

**USED BY IN THIS PROJECT**

[derivativeAverage\\_comp\\_linear\\_radial](#)

[Read exact source](#)

**LEVEL 8**

Project-wide support

**body\_neg**

MathlibAnnex.PluckerBody.body\_neg

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[body](#), [generators\\_neg](#)**USED BY IN THIS PROJECT**[normalizedGenerator\\_mem\\_of\\_spherelsometry](#)[Read exact source](#)**LEVEL 8**

Project-wide support

**generators\_isCompact**

MathlibAnnex.PluckerBody.isCompact\_generators

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[ballVolumeScaledMaximalMinors\\_continuous](#),  
[generators\\_eq\\_image\\_union](#)**USED BY IN THIS PROJECT**[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#), [derivativeAverage\\_mem\\_body](#),  
[derivativeGenerator\\_integrable\\_of\\_seminormLipschitz](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 8**

Project-wide support

**orientedSatelliteSupport\_configuration**

MathlibAnnex.PluckerSupport.orientedSatelliteSupport\_configuration

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[orientedSatelliteSupport](#),  
[pluckerFunctional\\_normalized\\_configurationFinMap](#)**USED BY IN THIS PROJECT**[orientedSatelliteSupport\\_self](#)[Read exact source](#)**LEVEL 8**

Project-wide support

**abs\_satelliteTerms\_le\_budget**

MathlibAnnex.Satellite.abs\_satelliteTerms\_le\_budget

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_replacementDeterminant\\_le\\_maximum](#), [satelliteBudget](#)**USED BY IN THIS PROJECT**[abs\\_satellitePolynomial\\_le](#)[Read exact source](#)

**LEVEL 8**

R01

**finiteDimensional\_codomain**

MathlibAnnex.Sphere.finiteDimensional\_codomain

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[radialExtensionHomeomorph](#)**USED BY IN THIS PROJECT**[finiteDimensional\\_domain](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_d  
omain](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 8**

Project-wide support

**radialExtension\_integral\_abs\_det**MathlibAnnex.Sphere.radialExtension\_integral\_abs  
\_det

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[volume\\_image\\_eq\\_zero\\_of\\_lipschitzWith](#),  
[closedUnitBallVolume](#), [dist\\_space\\_eq](#)**▼ Show 5 more**[closedUnitBall\\_isCompact](#),  
[orderedRows\\_ofOrderEmbedding](#),  
[integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#),  
[radialExtension\\_lipschitz](#), [radialExtension\\_image\\_ball](#)**USED BY IN THIS PROJECT**

None in this Project

[Open Declaration Card](#) [Read exact source](#)**LEVEL 8**

Project-wide support

**exists\_localMollification\_constant**MathlibAnnex.WeakGradient.exists\_localMollificat  
ion\_constant

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[inner\\_isOpen](#), [localMollification\\_contDiff](#),  
[localMollification\\_fderiv\\_eq\\_zero](#)**USED BY IN THIS PROJECT**[localMollification\\_eq\\_at\\_inner\\_points](#)[Read exact source](#)**LEVEL 8**

R02

**lexicographicRefine\_agreesOn**

MathlibAnnex.apply\_eq\_of\_mem\_lexicographicRefine

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[lexicographicRefine\\_subset](#)**USED BY IN THIS PROJECT**[lexicographicRefine\\_subsingleton](#)[Read exact source](#)**Level 9** 11 declarations

LEVEL 9

R07

## inverseBoundConstant\_pos

MathlibAnnex.DeterminantFrame.inverseBoundConstant\_pos

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[determinantMaximum\\_pos](#), [inverseBoundConstant](#)

USED BY IN THIS PROJECT  
[exists\\_nearMaxInverseBound](#)

[Read exact source](#)

LEVEL 9

R07

## nearMaxFrame\_det\_ne\_zero

MathlibAnnex.DeterminantFrame.nearMaxFrame\_det\_ne\_zero

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[nearMaxFrame\\_abs\\_det\\_lower](#)

USED BY IN THIS PROJECT  
[abs\\_inverse\\_mulVec\\_apply\\_le](#), [nearMaxFrame\\_det\\_isUnit](#),  
[absoluteMaximizer\\_satellite\\_norms\\_preimage](#)

[Read exact source](#)

LEVEL 9

R04

## integral\_det\_fderiv\_add\_sub\_eq\_zero\_of\_contDiff

MathlibAnnex.Piola.integral\_det\_fderiv\_add\_sub\_eq\_zero\_of\_contDiff

Measure or volume result

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[integral\\_det\\_singleOutputPerturb\\_sub\\_eq\\_zero](#),  
[outputHybrid\\_empty](#),  
[outputHybrid\\_insert\\_eq\\_singleOutputPerturb](#)

▼ Show 1 more  
[outputHybrid\\_univ](#)

USED BY IN THIS PROJECT  
[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 9

Project-wide support

## derivativeAverage\_mem\_body

MathlibAnnex.Plucker.derivativeAverage\_mem\_body

Determinant or plucker component

IMMEDIATE PREREQUISITES IN THIS PROJECT  
[closedUnitBallVolume\\_pos](#), [derivativeAverage](#),  
[derivativeGenerator\\_mem\\_raw](#)

▼ Show 3 more  
[body](#), [generators\\_isCompact](#), [isCompact\\_convexHull\\_pi](#)

USED BY IN THIS PROJECT  
[exists\\_extension\\_with\\_derivativeAverage\\_mem](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 9

Project-wide support

## derivativeGenerator\_integrable\_of\_semi\_normLipschitz

MathlibAnnex.Plucker.integrableOn\_derivativeGenerator\_of\_seminormLipschitz

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBall\\_isCompact](#),  
[ae\\_norm\\_apply\\_le\\_seminorm\\_of\\_lipschitz](#),  
[derivativeGenerator\\_mem\\_raw](#)

▼ Show 1 more

[generators\\_isCompact](#)

### USED BY IN THIS PROJECT

[exists\\_extension\\_with\\_derivativeAverage\\_mem](#)

[Read exact source](#)

LEVEL 9

Project-wide support

## orientedSatelliteSupport\_self

MathlibAnnex.PluckerSupport.orientedSatelliteSupport\_self

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[orientedSatelliteSupport\\_configuration](#)

### USED BY IN THIS PROJECT

[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#)

[Read exact source](#)

LEVEL 9

Project-wide support

## abs\_satellitePolynomial\_le

MathlibAnnex.Satellite.abs\_satellitePolynomial\_le

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[abs\\_satelliteTerms\\_le\\_budget](#), [satellitePolynomial](#)

### USED BY IN THIS PROJECT

[base\\_mem\\_nearMax\\_of\\_benchmark\\_le](#)

[Read exact source](#)

LEVEL 9

Project-wide support

## exists\_weight\_dominating\_budget

MathlibAnnex.Satellite.exists\_weight\_dominating\_budget

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[determinantMaximum\\_pos](#), [satelliteBudget](#)

### USED BY IN THIS PROJECT

[exists\\_goodNetSatelliteMaximizer](#)

[Read exact source](#)

LEVEL 9

R01

## finiteDimensional\_domain

MathlibAnnex.Sphere.finiteDimensional\_domain

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[finiteDimensional\\_codomain](#)

### USED BY IN THIS PROJECT

[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_codomain](#)

[Read exact source](#)

LEVEL 9

Project-wide support

## localMollification\_eq\_at\_inner\_points

MathlibAnnex.WeakGradient.localMollification\_eq\_at\_inner\_points

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists\\_localMollification\\_constant](#)

### USED BY IN THIS PROJECT

[compactLocalization\\_aeConstant\\_inner](#)

[Read exact source](#)

**LEVEL 9**

R02

**lexicographicRefine\_subsingleton**`MathlibAnnex.lexicographicRefine_subsingleton`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[lexicographicRefine\\_agreesOn](#)**USED BY IN THIS PROJECT**[exists\\_common\\_of\\_convexHull\\_eq](#)[Read exact source](#)**Level 10** 9 declarations**LEVEL 10**

R07

**abs\_inverse\_mulVec\_apply\_le**`MathlibAnnex.DeterminantFrame.abs_inverse_mulVec_apply_le`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_cramerNumerator\\_le](#), [coordinateFrame\\_mem](#), [coordinateRow\\_apply](#)**▼ Show 2 more**[nearMaxFrame\\_det\\_ne\\_zero](#), [cramerReplacement](#)**USED BY IN THIS PROJECT**[inverseBoundConstant\\_bound](#)[Read exact source](#)**LEVEL 10**

R07

**nearMaxFrame\_det\_isUnit**`MathlibAnnex.DeterminantFrame.isUnit_nearMaxFrame_det`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[nearMaxFrame\\_det\\_ne\\_zero](#)**USED BY IN THIS PROJECT**[inverse\\_mulVec\\_frameCoordinates](#)[Read exact source](#)

LEVEL 10

Project-wide support

## rawOrientedSupportMaximizer\_isAbsolutePolynomialMaximizer

MathlibAnnex.FiniteRecovery.exists\_contraction\_maximizing\_abs\_configurationPolynomial\_of\_mem\_maxSlice

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[closedUnitBallVolume\\_pos](#),  
[configurationFinMap\\_finMapConfiguration](#),  
[configuration\\_contraction\\_iff](#)

#### ▼ Show 6 more

[le\\_maximizer](#), [maxSlice](#), [generators\\_nonempty](#),  
[generators\\_isCompact](#), [orientedSatelliteSupport\\_self](#),  
[maxSatelliteConfiguration](#)

### USED BY IN THIS PROJECT

[targetRawSupportMaximizer\\_good](#)

[Read exact source](#)

LEVEL 10

Project-wide support

## integral\_maximalMinor\_fderiv\_add\_sub\_eq\_zero\_of\_contDiff

MathlibAnnex.NullLagrangian.integral\_maximalMinor\_fderiv\_add\_sub\_eq\_zero\_of\_contDiff

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[selectedOutput\\_hasCompactSupport](#),  
[maximalMinorIntegrand](#),  
[integral\\_det\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDiff](#)

### USED BY IN THIS PROJECT

[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)

[Read exact source](#)

LEVEL 10

Project-wide support

## exists\_extension\_with\_derivativeAverage\_mem

MathlibAnnex.Plucker.exists\_extension\_with\_derivativeAverage\_mem

Determinant or plucker component

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[dist\\_space\\_eq](#), [unitSphere](#), [derivativeAverage\\_mem\\_body](#)

#### ▼ Show 1 more

[derivativeGenerator\\_integrable\\_of\\_seminormLipschitz](#)

### USED BY IN THIS PROJECT

[normalizedGenerator\\_mem\\_of\\_spherelsometry](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 10

Project-wide support

## absoluteMaximizer\_satellite\_norms\_preimage

MathlibAnnex.Satellite.absoluteMaximizer\_satellite\_norms\_preimage

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[nearMaxFrame\\_det\\_ne\\_zero](#), [cramerReplacement](#), [Space](#)

#### ▼ Show 4 more

[eq\\_zero\\_of\\_apply\\_eq\\_zero](#), [abs\\_affine\\_endpoint\\_rigidity](#),  
[configurationPolynomial](#), [satelliteConfigurationSet](#)

### USED BY IN THIS PROJECT

[absoluteMaximizer\\_satellites\\_almost\\_norm](#)

[Open Declaration Card](#) [Read exact source](#)

**LEVEL 10**

Project-wide support

**base\_mem\_nearMax\_of\_benchmark\_le**`MathlibAnnex.Satellite.base_mem_nearMax_of_benchmark_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_satellitePolynomial\\_le](#)**USED BY IN THIS PROJECT**[absoluteMaximizer\\_base\\_nearMax](#)[Read exact source](#)**LEVEL 10**

Project-wide support

**compactLocalization\_aeConstant\_inner**`MathlibAnnex.WeakGradient.compactLocalization_aeConstant_inner`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[AEConstantOn](#), [exists\\_inner\\_convergencePoint](#),  
[localMollification\\_eq\\_at\\_inner\\_points](#)**USED BY IN THIS PROJECT**[exists\\_aeConstant\\_inner](#)[Read exact source](#)**LEVEL 10**

R02

**exists\_common\_of\_convexHull\_eq**`MathlibAnnex.exists_common_of_convexHull_eq`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[lexicographicRefine\\_convexHull\\_eq](#),  
[lexicographicRefine\\_subsingleton](#)**USED BY IN THIS PROJECT**[exists\\_common\\_pi](#)[Open Declaration Card](#) [Read exact source](#)**Level 11** 7 declarations**LEVEL 11**

R07

**inverseBoundConstant\_bound**`MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[abs\\_inverse\\_mulVec\\_apply\\_le](#), [determinantMaximum\\_pos](#),  
[inverseBoundConstant](#)**USED BY IN THIS PROJECT**[exists\\_nearMaxInverseBound](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 11**

R07

**inverse\_mulVec\_frameCoordinates**`MathlibAnnex.DeterminantFrame.inverse_mulVec_frameCoordinates`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[frameCoordinates\\_eq\\_mulVec](#), [nearMaxFrame\\_det\\_isUnit](#)**USED BY IN THIS PROJECT**[frameCoordinates\\_mem\\_coefficientAnnulus](#),  
[modelDist\\_framePreimage\\_le](#)[Read exact source](#)

LEVEL 11

Project-wide support

## integral\_maximalMinor\_fderiv\_add\_sub\_eq\_zero\_of\_lipschitzWith

MathlibAnnex.NullLagrangian.integral\_maximalMinor\_fderiv\_add\_sub\_eq\_zero\_of\_lipschitzWith

Measure or volume result

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[eventually\\_tsupport\\_mollify\\_subset\\_compact](#), [mollify\\_add](#), [maximalMinorIntegrand\\_mollify\\_continuous](#)

#### ▼ Show 4 more

[integrableOn\\_maximalMinor\\_fderiv\\_of\\_lipschitzWith](#), [integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_contDif](#), [f\\_integral\\_topMinor\\_difference\\_eq\\_setIntegral](#), [tendsto\\_integral\\_maximalMinor\\_mollify](#)

### USED BY IN THIS PROJECT

[weak\\_piola](#), [integral\\_maximalMinor\\_eq\\_of\\_pointwise\\_boundary\\_eq](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 11

Project-wide support

## absoluteMaximizer\_base\_nearMax

MathlibAnnex.Satellite.absoluteMaximizer\_base\_nearMax

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[frameMatrix\\_replaceRow](#), [maximizingFrame\\_mem](#), [base\\_mem\\_nearMax\\_of\\_benchmark\\_le](#)

#### ▼ Show 2 more

[configurationPolynomial](#), [satelliteConfigurationSet](#)

### USED BY IN THIS PROJECT

[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 11

Project-wide support

## absoluteMaximizer\_satellites\_almost\_norm

MathlibAnnex.Satellite.absoluteMaximizer\_satellites\_almost\_norm

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CoefficientDetectsUnit](#), [abs\\_eval\\_lower\\_of\\_norms\\_nearby](#), [absoluteMaximizer\\_satellite\\_norms\\_preimage](#)

### USED BY IN THIS PROJECT

[absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 11

Project-wide support

## exists\_aeConstant\_inner

MathlibAnnex.WeakGradient.exists\_aeConstant\_inner

Proposition or proof step

### IMMEDIATE PREREQUISITES IN THIS PROJECT

[compactLocalization\\_aeConstant\\_inner](#), [compactLocalization\\_eq\\_ae\\_inner](#)

### USED BY IN THIS PROJECT

[exists\\_localAEConstantAt](#)

[Read exact source](#)

## LEVEL 11

R02

### exists\_common\_pi

MathlibAnnex.exists\_common\_pi

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[allCoordinates\\_separate](#),  
[exists\\_common\\_of\\_convexHull\\_eq](#)

#### USED BY IN THIS PROJECT

[exists\\_almostIsometryMatch\\_at\\_satelliteDimension](#),  
[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

[Read exact source](#)

## Level 12 7 declarations

## LEVEL 12

Project-wide support

### weak\_piola

MathlibAnnex.BilipschitzOrientation.weak\_piola

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[f\\_injective](#), [contDiff](#), [fderiv\\_eq\\_zero\\_of\\_not\\_mem\\_carrier](#)

#### ▼ Show 5 more

[carrier\\_compact](#), [support\\_subset](#),  
[orderedRows\\_ofOrderEmbedding](#),  
[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#), [divergence\\_pi](#)

#### USED BY IN THIS PROJECT

[targetJacobianSign\\_weakDivergenceZero](#)

[Open Declaration Card](#) [Read exact source](#)

## LEVEL 12

R07

### exists\_nearMaxInverseBound

MathlibAnnex.DeterminantFrame.nonempty\_nearMaxInverseBound

Determinant or plucker component

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[NearMaxInverseBound](#), [inverseBoundConstant\\_bound](#),  
[inverseBoundConstant\\_pos](#)

#### USED BY IN THIS PROJECT

[exists\\_goodSatellitePackage](#)

[Read exact source](#)

**LEVEL 12**

Project-wide support

**integral\_maximalMinor\_eq\_of\_pointwise\_boundary\_eq**`MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq`

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[integral\\_maximalMinor\\_fderiv\\_add\\_sub\\_eq\\_zero\\_of\\_lipschitzWith](#)**USED BY IN THIS PROJECT**[normalizedGenerator\\_mem\\_of\\_spherisometry](#)[Open Declaration Card](#)[Read exact source](#)**LEVEL 12**

Project-wide support

**absoluteMaximizer\_satellites\_one\_sub\_epsilon**`MathlibAnnex.Satellite.absoluteMaximizer_satellites_one_sub_epsilon`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[absoluteMaximizer\\_satellites\\_almost\\_norm](#)**USED BY IN THIS PROJECT**[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#)[Read exact source](#)**LEVEL 12**

Project-wide support

**frameCoordinates\_mem\_coefficientAnnulus**`MathlibAnnex.Satellite.frameCoordinates_mem_coefficientAnnulus`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[NearMaxInverseBound](#), [inverse\\_mulVec\\_frameCoordinates\\_Space](#)**▼ Show 3 more**[eq\\_zero\\_of\\_apply\\_eq\\_zero](#), [coefficientAnnulus](#), [frameCoordinates\\_norm\\_le\\_model](#)**USED BY IN THIS PROJECT**[exists\\_net\\_preimage\\_close](#)[Read exact source](#)**LEVEL 12**

Project-wide support

**modelDist\_framePreimage\_le**`MathlibAnnex.Satellite.modelDist_framePreimage_le`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[NearMaxInverseBound](#), [inverse\\_mulVec\\_frameCoordinates\\_Space](#)**▼ Show 1 more**[eq\\_zero\\_of\\_apply\\_eq\\_zero](#)**USED BY IN THIS PROJECT**[exists\\_net\\_preimage\\_close](#)[Read exact source](#)

**LEVEL 12**

Project-wide support

**exists\_localAECConstantAt**

MathlibAnnex.WeakGradient.nonempty\_localAECConstantAt

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[LocalAECConstantAt](#), [exists\\_aeConstant\\_inner](#),  
[exists\\_localBallData](#)

**USED BY IN THIS PROJECT**

[locally\\_aeConstant](#)

[Read exact source](#)

**Level 13** 3 declarations**LEVEL 13**

Project-wide support

**targetJacobianSign\_weakDivergenceZero**

MathlibAnnex.BilipschitzOrientation.weakDivergenceZero\_targetJacobianSign

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[signed\\_area\\_transfer](#), [weak\\_piola](#), [WeakDivergenceZero](#)

**USED BY IN THIS PROJECT**

[targetJacobianSign\\_ae\\_const](#)

[Open Declaration Card](#) [Read exact source](#)

**LEVEL 13**

Project-wide support

**exists\_net\_preimage\_close**

MathlibAnnex.Satellite.exists\_net\_preimage\_close

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[K\\_nonneg](#), [FiniteCoefficientNet](#),  
[frameCoordinates\\_mem\\_coefficientAnnulus](#)

**▼ Show 1 more**

[modelDist\\_framePreimage\\_le](#)

**USED BY IN THIS PROJECT**

[finiteCoefficientNet\\_detectsUnit](#)

[Read exact source](#)

**LEVEL 13**

R05

**locally\_aeConstant**

MathlibAnnex.WeakDivergenceZero.nonempty\_localAECConstantAt

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[exists\\_localAECConstantAt](#)

**USED BY IN THIS PROJECT**

[exists\\_aeConstantOn](#)

[Open Declaration Card](#) [Read exact source](#)

## Level 14 2 declarations

### LEVEL 14

Project-wide support

#### finiteCoefficientNet\_detectsUnit

MathlibAnnex.Satellite.finiteCoefficientNet\_detectsUnit

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[CoefficientDetectsUnit](#), [exists\\_net\\_preimage\\_close](#)

#### USED BY IN THIS PROJECT

[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#)

[Read exact source](#)

### LEVEL 14

R05

#### exists\_aeConstantOn

MathlibAnnex.WeakDivergenceZero.exists\_aeConstantOn

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[locally\\_aeConstant](#),  
[exists\\_aeConstantOn\\_of\\_local\\_preconnected](#)

#### USED BY IN THIS PROJECT

[targetJacobianSign\\_ae\\_const](#), [exists\\_eqOn](#)

[Open Declaration Card](#) [Read exact source](#)

## Level 15 3 declarations

### LEVEL 15

Project-wide support

#### targetJacobianSign\_ae\_const

MathlibAnnex.BilipschitzOrientation.targetJacobianSign\_ae\_const

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[targetJacobianSign\\_locallyIntegrableOn](#),  
[targetJacobianSign\\_weakDivergenceZero](#),  
[exists\\_aeConstantOn](#)

#### USED BY IN THIS PROJECT

[integral\\_det\\_fderiv\\_eq\\_signed\\_volume](#)

[Open Declaration Card](#) [Read exact source](#)

### LEVEL 15

Project-wide support

#### finiteNet\_absoluteMaximizer\_satellites\_one\_sub\_epsilon

MathlibAnnex.Satellite.finiteNet\_absoluteMaximizer\_satellites\_one\_sub\_epsilon

Proposition or proof step

#### IMMEDIATE PREREQUISITES IN THIS PROJECT

[absoluteMaximizer\\_base\\_nearMax](#),  
[absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#),  
[finiteCoefficientNet\\_detectsUnit](#)

#### USED BY IN THIS PROJECT

[targetRawSupportMaximizer\\_good](#),  
[maxNetConfigurationMap\\_lower\\_of\\_ne\\_zero](#),  
[exists\\_goodNetSatelliteMaximizer](#)

[Read exact source](#)

**LEVEL 15**

R05

**exists\_eqOn**

MathlibAnnex.WeakDivergenceZero.exists\_eqOn

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[exists\\_aeConstantOn](#)**USED BY IN THIS PROJECT**

None in this Project

[Open Declaration Card](#) [Read exact source](#)**Level 16** 4 declarations**LEVEL 16**

Project-wide support

**integral\_det\_fderiv\_eq\_signed\_volume**

MathlibAnnex.BilipschitzOrientation.integral\_det\_fderiv\_eq\_signed\_volume

Measure or volume result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[targetJacobianSign\\_ae\\_const](#),  
[targetJacobianSign\\_const\\_is\\_pm\\_one](#)**USED BY IN THIS PROJECT**[derivativeAverage\\_comp\\_linear\\_radial](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 16**

Project-wide support

**targetRawSupportMaximizer\_good**

MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer\_good

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[PluckerGeneratorGood](#),  
[rawOrientedSupportMaximizer\\_isAbsolutePolynomialMaximizer](#),  
[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#)**USED BY IN THIS PROJECT**[exists\\_almostIsometryMatch\\_at\\_satelliteDimension](#),  
[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)[Open Declaration Card](#) [Read exact source](#)

**LEVEL 16**

Project-wide support

**maxNetConfigurationMap\_lower\_of\_ne\_zero**`MathlibAnnex.FiniteSup.maxNetConfigurationMap_lower_of_ne_zero`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[configurationMap\\_norm\\_gt\\_of\\_satellite](#),  
[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#),  
[maxSatelliteConfiguration](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**exists\_goodNetSatelliteMaximizer**`MathlibAnnex.Satellite.exists_goodNetSatelliteMaximizer`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[exists\\_weight\\_dominating\\_budget](#),  
[finiteNet\\_absoluteMaximizer\\_satellites\\_one\\_sub\\_epsilon](#),  
[maxSatelliteConfiguration](#)**USED BY IN THIS PROJECT**[exists\\_goodSatellitePackage](#)[Read exact source](#)**Level 17** 3 declarations**LEVEL 17**

Project-wide support

**exists\_almostIsometryMatch\_at\_satelliteDimension**`MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_at_satelliteDimension`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[AlmostIsometryMatch](#), [targetRawSupportMaximizer\\_good](#),  
[body](#)**▼ Show 1 more**[exists\\_common\\_pi](#)**USED BY IN THIS PROJECT**[exists\\_almostIsometryMatch\\_of\\_all\\_body\\_eq](#)[Read exact source](#)**LEVEL 17**

Project-wide support

**derivativeAverage\_comp\_linear\_radial**`MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[integral\\_det\\_fderiv\\_eq\\_signed\\_volume](#),  
[derivativeAverage\\_comp\\_linear\\_radial\\_apply](#),  
[radialExtension\\_leftInverse](#)**USED BY IN THIS PROJECT**[normalizedGenerator\\_mem\\_of\\_spherelsometry](#)[Open Declaration Card](#) [Read exact source](#)

**LEVEL 17**

Project-wide support

**exists\_goodSatellitePackage**`MathlibAnnex.Satellite.exists_goodSatellitePackage`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[exists\\_nearMaxInverseBound](#),  
[exists\\_goodNetSatelliteMaximizer](#)**USED BY IN THIS PROJECT**[exists\\_almostIsometryMatch\\_of\\_all\\_body\\_eq](#),  
[exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)[Open Declaration Card](#) [Read exact source](#)**Level 18** 3 declarations**LEVEL 18**

Project-wide support

**exists\_almostIsometryMatch\_of\_all\_body\_eq**`MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_of_all_body_eq`

Proposition or proof step

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[exists\\_almostIsometryMatch\\_at\\_satelliteDimension](#),  
[exists\\_goodSatellitePackage](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**normalizedGenerator\_mem\_of\_spherelsometry**`MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry`

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[integral\\_maximalMinor\\_eq\\_of\\_pointwise\\_boundary\\_eq](#),  
[derivativeAverage\\_comp\\_linear\\_radial](#),  
[exists\\_extension\\_with\\_derivativeAverage\\_mem](#)**▼ Show 2 more**[body\\_neg](#), [radialExtension\\_on\\_sphere](#)**USED BY IN THIS PROJECT**[rawGenerators\\_subset\\_of\\_spherelsometry](#)[Open Declaration Card](#) [Read exact source](#)

**LEVEL 18**

Project-wide support

**exists\_linearCertificate\_of\_pluckerBodies\_eq**

MathlibAnnex.PluckerRecovery.nonempty\_linearCertificate\_of\_pluckerBodies\_eq

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[targetRawSupportMaximizer\\_good](#), [lower\\_of\\_ne\\_zero](#), [maximalMinor\\_mul](#)

**▼ Show 6 more**

[mulVec\\_injective\\_of\\_maximalMinor\\_ne\\_zero](#), [mul\\_chartFactor\\_eq\\_of\\_maximalMinorsProportional\\_body](#), [LinearCertificate](#), [exists\\_goodSatellitePackage](#), [exists\\_common\\_pi](#)

**USED BY IN THIS PROJECT**

[exists\\_limitCertificate\\_of\\_pluckerBodies\\_eq](#)

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**Level 19** 2 declarations**LEVEL 19**

Project-wide support

**rawGenerators\_subset\_of\_spherelsometry**

MathlibAnnex.PluckerBody.rawGenerators\_subset\_of\_sphereIsometry

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[normalizedGenerator\\_mem\\_of\\_spherelsometry](#)

**USED BY IN THIS PROJECT**

[subset\\_of\\_spherelsometry](#)

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**LEVEL 19**

Project-wide support

**exists\_limitCertificate\_of\_pluckerBodies\_eq**

MathlibAnnex.PluckerRecovery.nonempty\_limitCertificate\_of\_pluckerBodies\_eq

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[LimitCertificate](#), [referenceNorm\\_le](#), [exists\\_linearCertificate\\_of\\_pluckerBodies\\_eq](#)

**USED BY IN THIS PROJECT**

[exists\\_linearsometryEquiv\\_of\\_pluckerBodies\\_eq](#)

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**Level 20** 2 declarations

**LEVEL 20**

Project-wide support

**exists\_linearIsometryEquiv\_of\_pluckerBodies\_eq**

MathlibAnnex.EquivalentSeminorm.nonempty\_linearIsometryEquiv\_of\_pluckerBodies\_eq

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[LinearIsometryEquiv](#), [image\\_unitBall\\_eq](#),  
[exists\\_limitCertificate\\_of\\_pluckerBodies\\_eq](#)

▼ Show 1 more

[map\\_eq](#)

**USED BY IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_spherIsometryEquiv](#)

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**LEVEL 20**

Project-wide support

**subset\_of\_spherIsometry**

MathlibAnnex.PluckerBody.subset\_of\_sphereIsometry

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[rawGenerators\\_subset\\_of\\_spherIsometry](#)

**USED BY IN THIS PROJECT**

[eq\\_of\\_spherIsometry](#)

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**Level 21** 1 declaration**LEVEL 21**

Project-wide support

**eq\_of\_spherIsometry**

MathlibAnnex.PluckerBody.eq\_of\_sphereIsometry

Determinant or plucker component

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[subset\\_of\\_spherIsometry](#)

**USED BY IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_spherIsometryEquiv](#)

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**Level 22** 1 declaration

**LEVEL 22**

Project-wide support

**linearIsometryEquiv\_of\_spherIsometryEquiv**

MathlibAnnex.EquivalentSeminorm.nonempty\_linearIsometryEquiv\_of\_sphereIsometryEquiv

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[exists\\_linearIsometryEquiv\\_of\\_pluckerBodies\\_eq](#),  
[eq\\_of\\_spherIsometry](#)

**USED BY IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_coordinate\\_model](#)

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**Level 23** 1 declaration**LEVEL 23**

Project-wide support

**linearIsometryEquiv\_of\_coordinate\_model**

MathlibAnnex.Sphere.nonempty\_linearIsometryEquiv\_of\_coordinate\_model

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_spherIsometryEquiv](#),  
[spherIsometryEquiv](#), [transportLinearIsometryEquiv](#)

**USED BY IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)

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**Level 24** 1 declaration

**LEVEL 24**

Project-wide support

**linearIsometryEquiv\_of\_isometryEquiv\_finiteDimensional**`MathlibAnnex.Sphere.nonempty_linearIsometryEquiv_of_isometryEquiv_finiteDimensional`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[finrank\\_eq](#), [linearIsometryEquiv\\_of\\_coordinate\\_model](#)**USED BY IN THIS PROJECT**[nonempty\\_isometryEquiv\\_iff\\_nonempty\\_linearIsometryEquiv\\_finiteDimensional](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_codomain](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_domain](#)**▼ Show 1 more**[linearIsometryEquiv\\_of\\_isometry\\_surjective\\_finiteDimensional](#)[Read exact source](#)**Level 25** 4 declarations**LEVEL 25**

Project-wide support

**nonempty\_isometryEquiv\_iff\_nonempty\_linearIsometryEquiv\_finiteDimensional**`MathlibAnnex.Sphere.nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_finiteDimensional`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[isometryEquiv\\_of\\_linearIsometryEquiv](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)**USED BY IN THIS PROJECT**

None in this Project

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Project-wide support

**linearIsometryEquiv\_of\_isometryEquiv\_finiteDimensional\_codomain**`MathlibAnnex.Sphere.nonempty_linearIsometryEquiv_of_isometryEquiv_finiteDimensional_codomain`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[finiteDimensional\\_domain](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)**USED BY IN THIS PROJECT**[linearIsometryEquiv\\_of\\_isometryEquiv\\_of\\_finiteDimensional](#)[Read exact source](#)

**LEVEL 25**

Project-wide support

**linearIsometryEquiv\_of\_isometryEquiv\_finiteDimensional\_domain**`MathlibAnnex.Sphere.nonempty_linearIsometryEquiv_of_isometryEquiv_finiteDimensional_domain`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[finiteDimensional\\_codomain,](#)  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)**USED BY IN THIS PROJECT**[linearIsometryEquiv\\_of\\_isometryEquiv\\_of\\_finiteDimensional](#)[Read exact source](#)**LEVEL 25**

Project-wide support

**linearIsometryEquiv\_of\_isometry\_surjective\_finiteDimensional**`MathlibAnnex.Sphere.nonempty_linearIsometryEquiv_of_isometry_surjective_finiteDimensional`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional](#)**USED BY IN THIS PROJECT**

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Project-wide support

**linearIsometryEquiv\_of\_isometryEquiv\_of\_finiteDimensional**`MathlibAnnex.Sphere.nonempty_linearIsometryEquiv_of_isometryEquiv_of_finiteDimensional`

Rigidity or equivalence result

**IMMEDIATE PREREQUISITES IN THIS PROJECT**[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_codomain,](#)  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_finiteDimensional\\_domain](#)**USED BY IN THIS PROJECT**[nonempty\\_isometryEquiv\\_iff\\_nonempty\\_linearIsometryEquiv\\_of\\_finiteDimensional,](#)  
[linearIsometryEquiv\\_of\\_isometry\\_surjective\\_of\\_finiteDimensional](#)[Read exact source](#)**Level 27** 2 declarations

LEVEL 27

Project-wide support

## **nonempty\_isometryEquiv\_iff\_nonempty\_linearIsometryEquiv\_of\_finiteDimensional**

MathlibAnnex.Sphere.nonempty\_isometryEquiv\_iff\_nonempty\_linearIsometryEquiv\_of\_finiteDimensional

Rigidity or equivalence result

### **IMMEDIATE PREREQUISITES IN THIS PROJECT**

[isometryEquiv\\_of\\_linearIsometryEquiv](#),  
[linearIsometryEquiv\\_of\\_isometryEquiv\\_of\\_finiteDimensional](#)

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Project-wide support

## **linearIsometryEquiv\_of\_isometry\_surjective\_of\_finiteDimensional**

MathlibAnnex.Sphere.nonempty\_linearIsometryEquiv\_of\_isometry\_surjective\_of\_finiteDimensional

Rigidity or equivalence result

### **IMMEDIATE PREREQUISITES IN THIS PROJECT**

[linearIsometryEquiv\\_of\\_isometryEquiv\\_of\\_finiteDimensional](#)

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